

Design of a 3.5MW Wind Turbine

WIND TURBINE DESIGN

AE4W09

Authors: Group 2



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Turbine Description:

3-Bladed, 3.5-MW, Onshore Wind Turbine, IEC Class II-B

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Contents

1	Market Selection and Outline	6
2	System Design	8
2.1	Summary of the design	8
2.2	Supporting material, analyses and rationale	9
2.2.1	Site Conditions and Operational range	9
2.2.2	Use of NREL reference and scaling	11
2.2.3	Levelized Production Cost optimization	11
2.2.4	Initial Design Parameters	13
3	Rotor Aerodynamic and Structural Design	16
3.1	Summary of the design	16
3.2	Supporting material, analyses and rationale	17
3.2.1	Airfoil selection and distribution	17
3.2.2	Chord and Twist Distribution Design	18
3.2.3	Structural Properties and Scaling Laws	21
3.2.4	Tip Deflection and Stresses	23
3.2.5	Overhang, Pre-bend, Tilt- and Cone-Angle	24
4	Drive Train Design	25
4.1	Drive train and electrical system configuration.	25
4.1.1	Summary of the design	25
4.1.2	Supporting material, analyses and rationale	25
4.2	Generator	29
4.2.1	Summary of the design	29
4.2.2	Supporting material, analyses and rationale	29
4.3	Gearbox	30
4.3.1	Summary of the design	30
4.3.2	Supporting material, analyses and rationale	30
4.4	Converter	31
4.4.1	Summary of the design	31

4.4.2	Supporting material, analyses and rationale	31
4.5	Power Cables in Tower	32
4.5.1	Summary of the design	32
4.5.2	Supporting material, analyses and rationale	32
4.6	Nacelle and Drivetrain Properties	32
4.6.1	Summary of the design	32
4.6.2	Supporting material, analyses and rationale	33
5	Steady State and Dynamics	36
5.1	Summary of the outcome and updates	36
5.2	Supporting material, analyses and rationale	38
5.2.1	Torque control	38
5.2.2	Performance analysis	40
5.2.3	Tower design	41
6	Control Design	45
6.1	Description of the control system	45
6.2	Controller tuning	47
6.3	Performance evaluation of the controller	48
7	Certification Analysis	52
7.1	Summary of the Analysis	52
7.2	Detailed Analyses and Interpretation	54
7.2.1	Turbine IEC Class Determination	54
7.2.2	Moments to Stresses	54
7.2.3	Turbine Behavior under Certification Load Cases	56
7.2.4	Selection of Partial Safety Factors for ULS and FLS	58
7.2.5	Ultimate Limit State Analysis	59
7.2.6	Fatigue Limit State Analysis	59
7.2.7	Mitigation of Tip Clearance Safety Factor	60
8	Reflection and Prospect	62
8.1	Reflection	62
8.2	Prospect	63

Abbreviations

AEP	Annual Energy Production
BEM	Blade-Element Momentum
CAPEX	Capital Expenditure
DLC	Design Load Case
DFIG	Doubly-Fed Induction Generator
DU	Delft University
ETM	Extreme turbulence model
EWM	Extreme wind model
EWS	Extreme wind shear
F	Fatigue load analysis
FLS	Fatigue Limit State
GFC	Full-scale Converter
IEC	International Electrotechnical Commission
LPC	Levelized Production Cost
N	Normal load analysis
NTM	Normal turbulence model
OPEX	Operational Expenditure
TSR	Tip-Speed Ratio
U	Ultimate load analysis
ULS	Ultimate Limit State
WTG	Wind Turbine Generator

Nomenclature

A	Weibull scale parameter [m/s]
$A_{cross-section}$	Cross-section area [m ²]
c	Chord length [m]
C	Blade tip clearance [m]
$C(D)$	Normalized cost as function of rotor diameter [-]
C_d	Drag coefficient [-]
C_l	Lift coefficient [-]
C_P	Power coefficient [-]
d_j	Damage for the given wind speed [-]
D	diameter [m]
D	Total damage
E	Elastic modulus [Pa]
E_y	Annual energy production [Wh/yr]
f	Grid frequency [Hz]

$f(v)$	Weibull probability density function
f_k	Ultimate Compressive Strength [-]
F_k	Stress result from FASTTool [-]
F_{thrust}	Thrust force on rotor [N]
GM	Gain margin [dB]
GR	Gearbox ratio [-]
h_{hub}	Hub height [m]
I	Second moment of area [m ⁴]
$I_{\text{Drivetrain}}$	Equivalent drivetrain inertia [kg·m ²]
I_{Gen}	Generator inertia [kg·m ²]
I_{Rotor}	Rotor inertia [kg·m ²]
K	Damping gain [-]
K_i	Integral gain [-]
K_{opt}	Optimal mode gain [-]
K_p	Proportional gain [-]
LPC	Levelized Production Cost [/Wh]
M	Bending moment [Nm]
m	Blade mass [kg]
m'	Distributed mass per length [kg/m]
Q	Shaft torque [Nm]
R	Rotor radius [m]
r	Radial position on the blade
SF	Safety factor [-]
t_{base}	Tower base wall thickness [m]
T_{Aero}	Aerodynamic torque [Nm]
T_{Gen}	Generator torque [Nm]
v	Wind speed [m/s]
$v_{\text{cut-in}}$	Cut-in wind speed [m/s]
$v_{\text{cut-out}}$	Cut-out wind speed [m/s]
v_{rated}	Rated wind speed [m/s]
V_{tip}	Tip speed of the blade [m/s]
x	Converter rating [-]
y_{tip}	Blade tip deflection [m]
α	Angle of attack [-]
β	Blade collective pitch angle [deg]
β	Blade pitch angle from pitch controller [deg]
β_1, β_2	First notch filter parameters [-]
$\Delta\beta$	Increase in pitch angle due to tower damper [deg]
δ_{max}	Maximum tip deflection [m]
γ_f	Load factor [-]

γ_m	Material factor [-]
γ_n	Consequence of failure factor [-]
γ_1, γ_2	Second notch filter parameters [-]
λ	Tip-speed ratio [-]
ω	Rotational speed [rad/s]
$\omega_1, \omega_2, \omega_{filter}$	Filter cut-off frequencies [rad/s]
ω_{gc}	Gain cross-over frequency [rad/s]
ρ	Air density [kg/m ³]
θ	Blade twist angle [deg]
σ	Bending stress [Pa]

Chapter 1

Market Selection and Outline

The global wind energy market size is valued at USD 87.66 billion in 2023 and is expected to grow to USD 190.39 billion in 2032 [4]. Despite the fact that the biggest portion of wind energy is generated by China, Europe comes third in the race, as seen in Figure 1.1, amounting to 28% of the global market share. This implies that there is substantial potential in further development of new turbines with better performance characteristics for the European market.

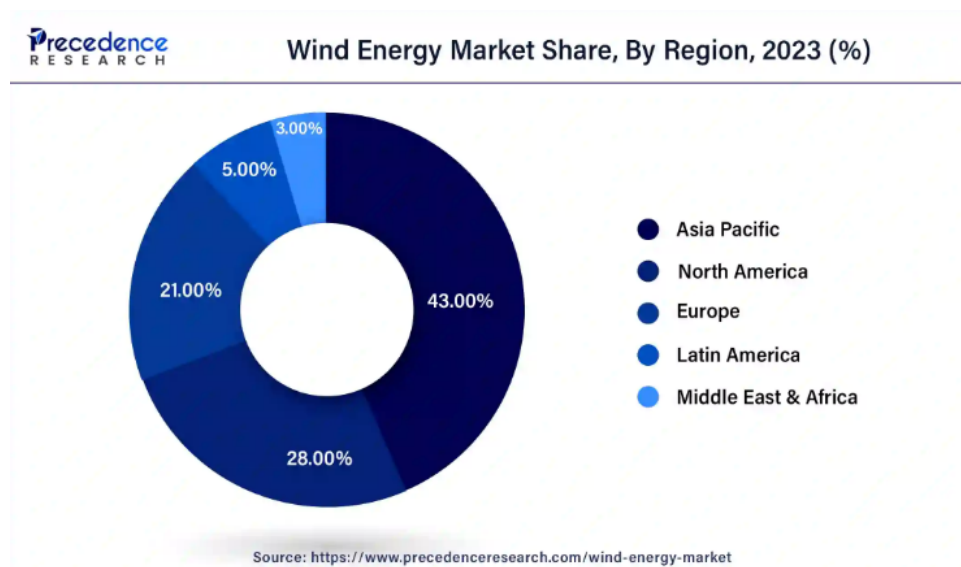


Figure 1.1: *Wind energy market share by region in 2023*

Moreover, looking at the current data and prognosis for the coming years, it is expected that between 2024-2030, two-thirds of all installations in Europe are going to be onshore [3]. In 2023, onshore installations accounted for 79% and with the alleviation of EU rules permitting onshore installations, the volume installed is expected to amount to 17% of the total electricity generated in Europe [3]. What is more, despite a favorable prognosis for offshore installations, it is still evident that onshore will play an important role, as it can be noticed from Figure 1.2. Finally, decommissioning old onshore turbines and replacing them

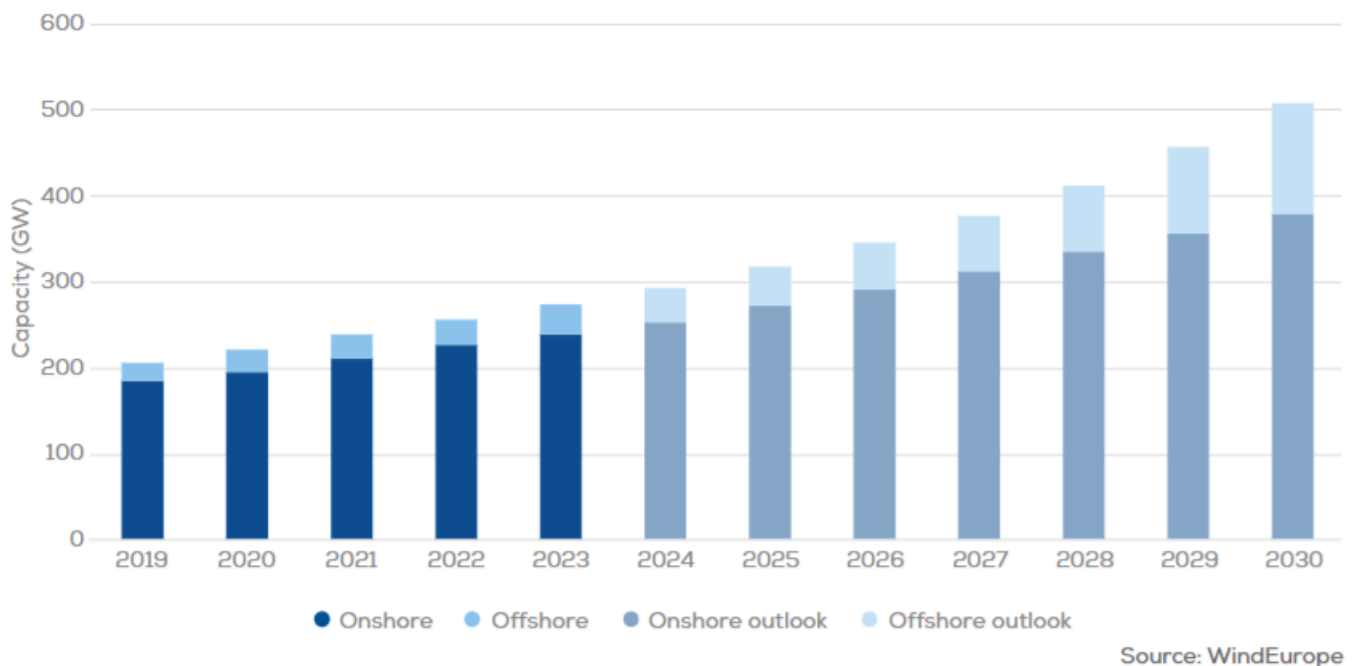


Figure 1.2: *New wind power in Europe - WindEurope's Outlook [3]*

with higher power rated ones is expected to reduce the total number of turbines by 25% while maintaining the same energy generation.

With that being said, it can be concluded that there is further potential to develop a high-performance wind turbine suitable for onshore locations. In years 2005 - 2014, the highest portion of installed turbines belonged to IEC class II A/B [14]. Therefore, IEC class II A/B is chosen for the design as it is a reflection of market need and allows to take advantage of the high wind speeds in coastal areas. Taking into consideration possible urban areas in such locations and noise regulations, the power rating of 3.5MW is chosen. This power rating is close to the average one installed in France (3.6MW), Poland (3.2MW), Spain (3.8MW) and Ireland (3.4MW) among other European countries in 2023 [3] which further justifies its business feasibility.

The most industry-grounded configuration solution is a Danish concept consisting of a three-bladed upwind rotor. By considering current turbine designs from companies which take up a large market share, such as Siemens Gamesa and Vestas, this is also the only configuration they offer. Therefore, this configuration is selected for the design turbine.

Chapter 2

System Design

2.1 Summary of the design

Table 2.1: *Objectives and boundary conditions.*

Parameter	Value / Description
Onshore or offshore	Onshore
Type of location:	Coastal area (Low Roughness length)
Wind conditions: Weibull scale factor	$A = 8.97$
Wind conditions: Weibull shape factor	$k = 2.24$
Wind conditions: reference height for Weibull parameters	100 [m]
Noise constraint	Maximum tip speed < 80[m/s]

Table 2.2: *Design variables / choices*

Parameter	Value / Description
Rated power	3.5 [MW]
Number of blades	3
Rotor diameter	117.5 [m]
Design tip speed ratio	$\lambda = 7$
Rotational speed range	3.47 - 12.04 [rpm]
Cut-in wind speed	3 [m/s]
Cut-out wind speed	25 [m/s]
Max low-speed shaft torque	2.77 [MNm]
Hub height	84 [m]

2.2 Supporting material, analyses and rationale

This section covers the high level decisions undertaken in the design process of the wind turbine. Firstly, representative sites are chosen allowing for establishing Weibull curve distribution of wind speeds in [subsection 2.2.1](#). Then, [subsection 2.2.2](#) describes the geometric scaling of reference NREL turbine to the design turbine. Through economic analysis of levelized production cost, the optimal turbine diameter is determined in [subsection 2.2.3](#). Finally, [subsection 2.2.4](#) gives estimated for initial design parameters.

2.2.1 Site Conditions and Operational range

This Wind Turbine is designed to perform well for a variety of onshore coastal locations in Europe. To represent these wind conditions and ensure the diverse range of wind conditions characteristic of IEC Class II wind turbines, five representative sites were selected across Europe. The geographical locations of these reference sites are shown in [Figure 2.1](#), including the following sites:

- **Denmark** (55.22222, 14.80556)
- **Poland** (54.13889, 15.44444)
- **Germany** (53.58333, 8.055555)
- **Scotland, UK** (54.91667, -4.75)
- **Netherlands** (51.75, 4.11111)

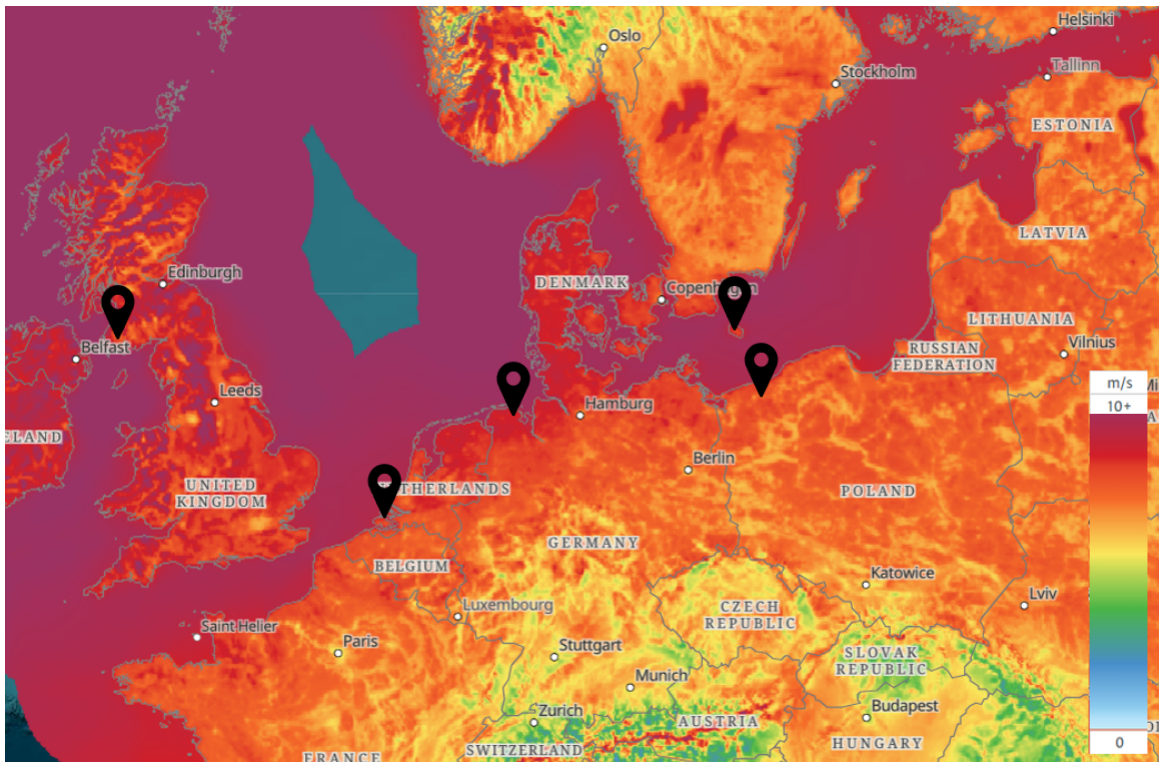


Figure 2.1: *Reference sites selected for representative wind conditions based on market analysis.*

For each site, a representative 3×3 [km] area containing historical wind speed data is used to depict the expected working conditions for the wind turbine. To model wind speed probability at each location, the Weibull distribution, given in Equation 2.1, provides an effective statistical representation of wind resource variability.

$$f(v) = \frac{k}{A} \left(\frac{v}{A} \right)^{k-1} e^{-\left(\frac{v}{A} \right)^k} \quad (2.1)$$

where A is the scale parameter [m/s], which defines the characteristic wind speed, k is the shape parameter, which determines the distribution's skewness, and v is the wind speed [m/s]. These parameters determine the frequency of different wind speeds, which directly influence the expected power production. The derived Weibull distributions play a crucial role in weighting the power output $P(v)$ when computing the Annual Energy Production (AEP) in the upcoming section.

To gather scale and shape parameters A and k , the *WaSP* software [26] extracted the Weibull distributions for the given sites from [12] for a range of wind speeds. The Weibull distributions for the chosen sites are depicted in Figure 2.2.

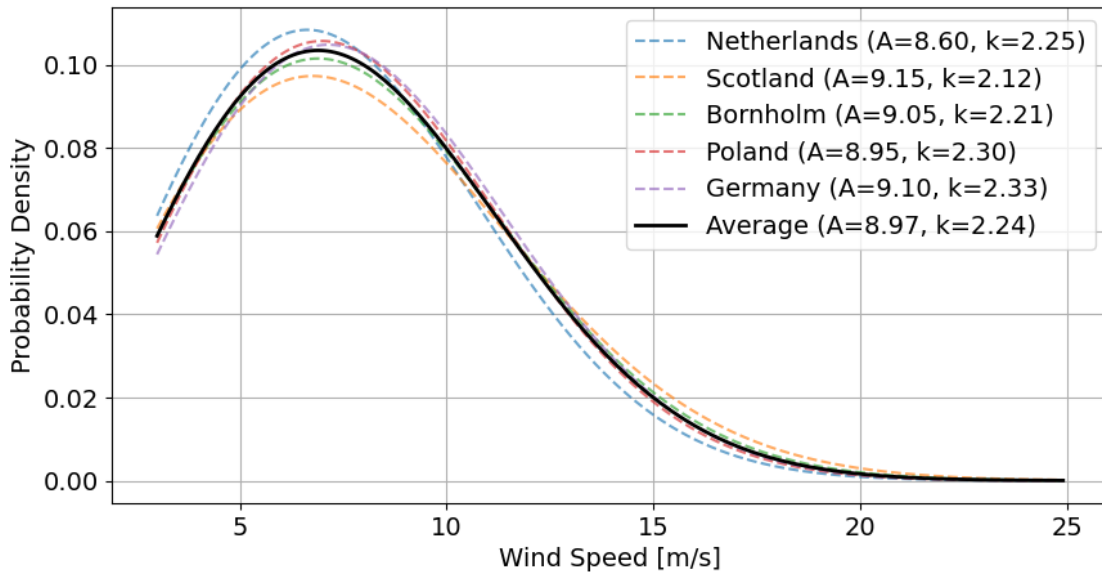


Figure 2.2: Weibull wind speed distributions for the selected locations, including the Netherlands, Scotland, Bornholm, Poland, and Germany. The shape (k) and scale (A) parameters for each site are provided in the legend. The average distribution, shown in black, represents the mean wind conditions across all locations.

From these distributions, an average wind speed distribution is determined, yielding the parameters $A = 8.97$ and $k = 2.24$. Using these averaged parameters in Equation 2.1 provides a reasonable representation of the wind conditions that can be expected at the target sites.

2.2.2 Use of NREL reference and scaling

Throughout this report, the 5MW NREL turbine [19] serves as a reference to establish adequate scaling of the design to ensure cost-effective energy production while maintaining performance consistency. The NREL 5 MW reference wind turbine has a rotor diameter of $D_{\text{ref}} = 126$ [m] and with the assumption of a constant rotor power loading at the rated wind speed, the rotor diameter is scaled for the 3.5 MW turbine as follows:

$$D_{\text{scale}} = D_{\text{ref}} \sqrt{\frac{P_{\text{rated}}}{P_{\text{ref}}}} = 126 \text{ [m]} \cdot \sqrt{\frac{3.5 \text{ [MW]}}{5.0 \text{ [MW]}}} \approx 105.4 \text{ m} \quad (2.2)$$

To evaluate the Annual Energy Production (AEP), a parametric power curve $P(v)$ dependent on wind speed v , constrained by the rated power limit $P_{\text{rated}} = 3.5$ MW, is defined as:

$$P(v) = \begin{cases} 0, & v < v_{\text{cut-in}} \text{ or } v > v_{\text{cut-out}} \\ \frac{1}{2} \rho \pi \left(\frac{D}{2}\right)^2 C_P v^3, & v_{\text{cut-in}} \leq v < v_{\text{rated}}, \text{ if } P(v) < P_{\text{rated}} \\ P_{\text{rated}}, & v \geq v_{\text{rated}} \end{cases} \quad (2.3)$$

where $C_p = 0.482$, is taken from the NREL 5 MW reference turbine [19].

2.2.3 Levelized Production Cost optimization

Under the linear geometric scaling assumption, the wind turbine's sweep area is ultimately responsible for the energy yield scales quadratically with radius $A \propto R^2$. However, the mass which is the largest contributor to the costs of the turbine experiences a cubic growth $M \propto R^3$ meaning the costs increase faster than energy yield with size.

To address this issue the approach to scaling the wind turbine is based on the normalized cost function $C(D)$ for an **onshore** wind turbine is modelled through [Equation 2.4](#).

$$C(D) = 0.70 + 0.30 \left(\frac{D}{D_{\text{scale}}} \right)^{2.6} \quad (2.4)$$

This equation accounts for the increase in capital and operational costs as the rotor diameter deviates from the scaled reference value. Combining the normalized cost function with the AEP given in [Equation 2.5](#) yields the Levelized Production Cost (LPC).

$$E_y = \int_{v_{\text{cut-in}}}^{v_{\text{cut-out}}} P(v) f(v) dv \quad (2.5)$$

The LPC Equation 2.6 is the ratio of the total costs and total energy output making it a useful metric for determining the most cost-effective rotor diameter by minimizing production costs while maximizing energy output.

$$\text{LPC} = \frac{C(D)}{E_y} \quad (2.6)$$

The power output for varying rotor diameters is computed and compared, determining an optimal design balancing cost and energy yield. Figure 2.3 illustrates how the Levelized Production Cost (LPC) and Annual Energy Production (AEP) vary with rotor diameter. The minimum LPC is achieved at an optimal diameter of **117.5 [m]**, where the trade-off between increased energy capture and structural/economic costs is most favourable.

As the rotor diameter increases:

- The **LPC** initially decreases due to higher energy capture efficiency but begins to rise again due to increased material and operational costs.
- The **AEP** follows a monotonically increasing trend, as larger rotors sweep a greater wind area, thus extracting more energy.

This balance between cost and energy yield is visually captured in Figure 2.3, where the intersection of the declining LPC curve and the rising AEP curve determines the optimal rotor diameter.

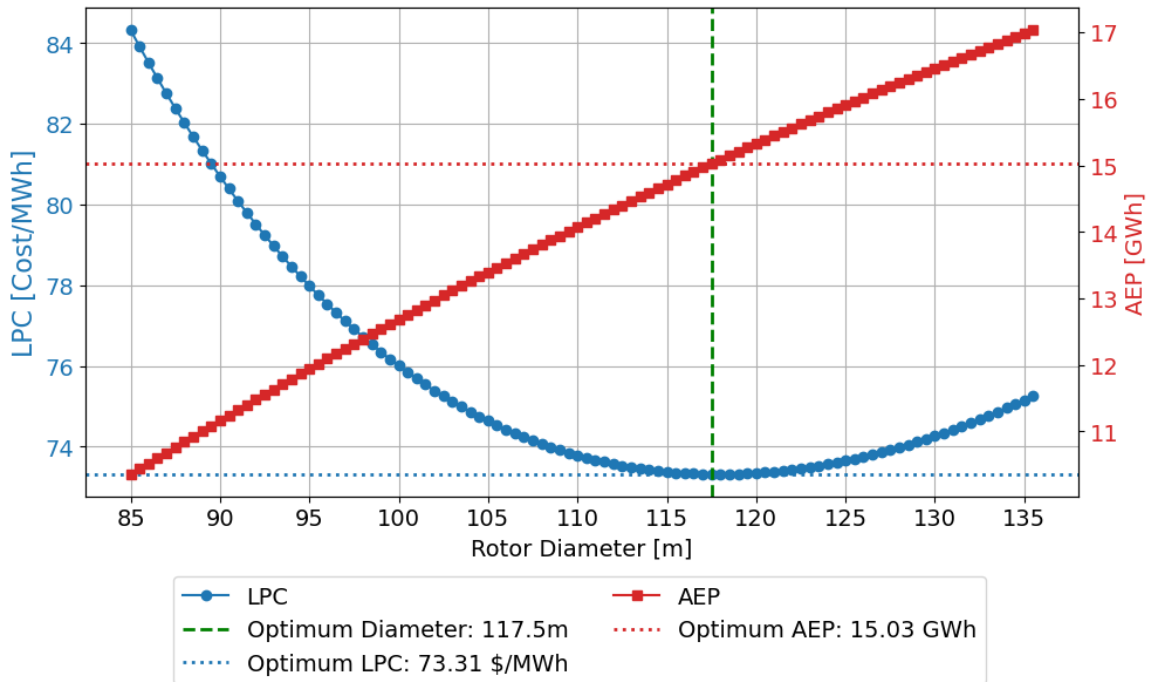


Figure 2.3: *Levelized Cost of Energy (LPC) and Annual Energy Production (AEP) versus rotor diameter. The minimum LPC is achieved at $D = 117.5$ [m], where economic feasibility is maximized.*

The impact of this optimal diameter on turbine performance is further analyzed in Figure 2.4, which presents the power curves for different rotor sizes. Notably, **the optimum diameter** leads to a rated wind speed (v_{rated}) of **10.4 [m/s]**, as compared to the reference NREL 3.5 MW turbine, which has a rated speed of 11.4 [m/s]. This indicates that a larger rotor reaches peak power at a lower wind speed, which can be advantageous in regions with moderate wind resources.

These findings indicate that a rotor diameter of 117.5 [m] provides the optimal balance between cost-efficiency and energy production, making it the most suitable design choice under the given economic and environmental conditions. At this diameter, the Levelized Production Cost (LPC) reaches its minimum value of 73.31 [\$/MWh], while the Annual Energy Production (AEP) is maximized at 15.03 [GWh]. This result confirms that increasing the rotor diameter beyond this point would lead to higher costs without proportional energy gains, while a smaller diameter would reduce energy capture and efficiency.

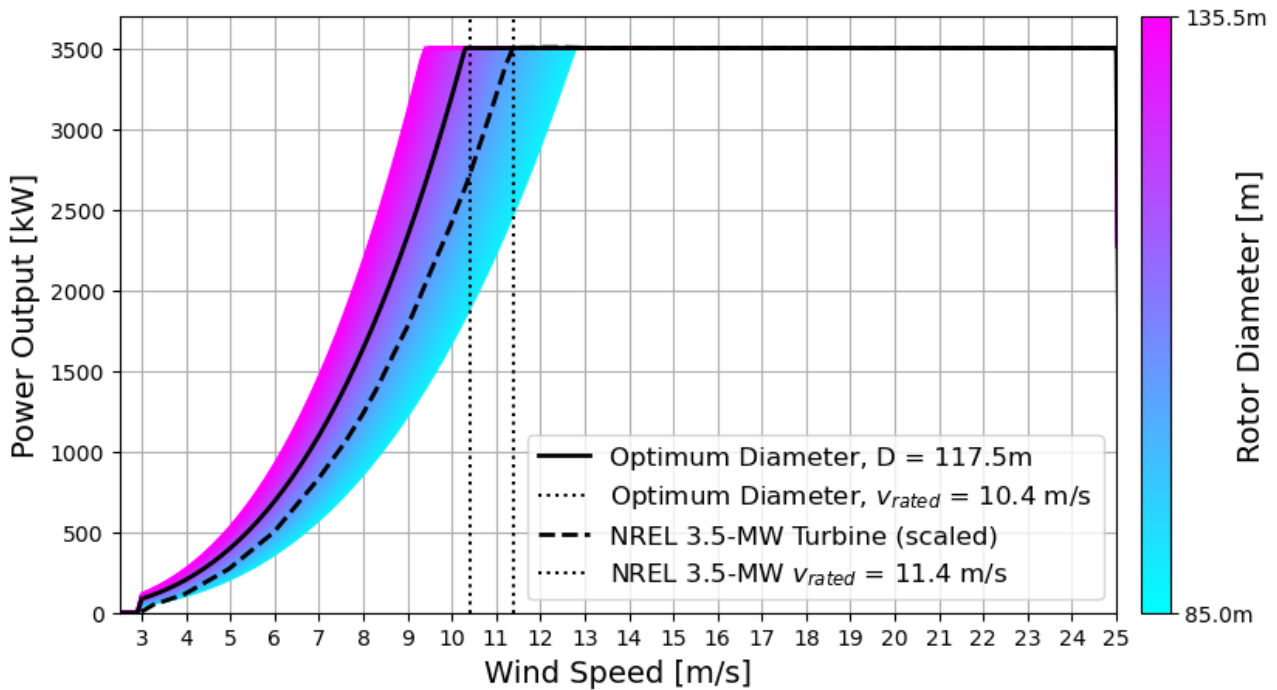


Figure 2.4: Power curves for different rotor diameters, highlighting the optimal diameter found through the LPC analysis. The shaded region represents variations in turbine performance as a function of rotor size.

2.2.4 Initial Design Parameters

Design tip speed ratio

With an established rotor diameter and rated wind speed, the remaining system-level design parameters are determined by selecting an adequate design tip speed ratio λ , Equation 2.7.

$$\lambda = \frac{\omega \cdot R}{V} \quad (2.7)$$

By fixing the tip-speed ratio, the corresponding rotational speed range ω can be plotted as a function of the wind speed as shown in Figure 2.5 for a $\lambda = 7$. Then speed at the tip of the rotor V_{tip} corresponds to $V_{tip} = \omega_{max}R$ and indicates the noise pollution of the wind turbine. Considering the location is onshore, staying below the limit of $V_{tip} = 80$ [m/s] is important for people who may reside in the area of the wind turbine. For this reason, the design tip-speed ratio $\lambda = 7$ has been selected to stay below the limit. By assuming that the hub width is zero $R \approx D/2$ the speed at the tip of the rotor is found to be:

$$V_{tip} = \omega_{max}R \approx 74 \text{ [m/s]}$$

Aside from choosing the design tip speed ratio, the following conclusions can be drawn from Figure 2.5. Firstly, the optimized rotor diameter exceeds the linearly scaled one, leading to a reduced maximum rotational speed. Secondly, due to the decreased rated wind speed, the turbine operates at its maximum rotational speed over a wider range of wind speeds. The numerical range of rotational speeds is summarized alongside other key turbine parameters in Table 2.2.

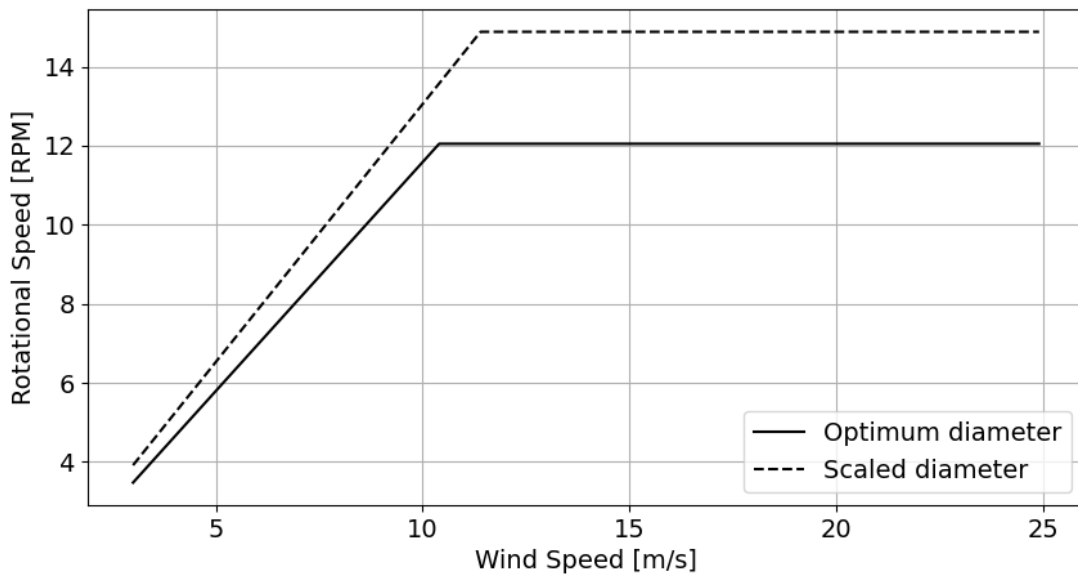


Figure 2.5: Rotational speed as a function of wind speed for both the scaled and optimized rotor diameters.

For the drivetrain design, a critical parameter is the low-speed shaft torque Q , which can be estimated from the rated power P_{rated} and rotational speed ω as:

$$Q = \frac{P}{\omega} = 2.77 \text{ [MNm]}. \quad (2.8)$$

Hub Height

The hub height of the redesigned turbine is determined by applying similarity scaling to the NREL 5 MW reference turbine, which has a hub height of 90 [m] [19]. Using first-order linear geometric scaling based on rotor diameter, the scaled hub height is computed as:

$$\text{Hub height} = 90 \text{ [m]} \cdot \left(\frac{117.7 \text{ [m]}}{126 \text{ [m]}} \right) = 84 \text{ [m]} \quad (2.9)$$

This approach preserves proportionality in structural layout while adjusting for the reduced rotor diameter of the new design.

Chapter 3

Rotor Aerodynamic and Structural Design

3.1 Summary of the design

Table 3.1: *Objectives and boundary conditions*

Parameter	Value / Description
Number of blades	3
Rotor diameter	117.5 [m]
Design tip speed ratio	7

Table 3.2: *Design variables and choices*

Parameter	Value / Description
Twist offset at tip (= blade pitch angle for optimal operation)	-
'Thickness factor' of the blade spar, web and skin	1.0
Overhang distance of rotor	5 [m]
Tilt angle of main shaft	3 [°]
Cone angle of rotor	2.5 [°] (upwind)
Pre-bending of blades	None
Optimal tip speed ratio	7.6 [-]
Optimal fine pitch angle	0 [°]

3.2 Supporting material, analyses and rationale

3.2.1 Airfoil selection and distribution

For large wind turbine blades, a combination of airfoil families is typically employed to balance aerodynamic efficiency, structural robustness, and manufacturability. Cylindrical sections (*Cylinder 1* and *Cylinder 2*) are used inboard, where the blade experiences high bending moments and must accommodate a secure root connection. Moving outward, thicker airfoils from the DU series (*DU 99-W-405*, *DU 99-W-350*, etc.) offer higher lift coefficients, favourable stall characteristics, and most importantly structural stiffness where the blade is still relatively thick to account for the connection to the cylinder. Finally, the tip regions use thinner profiles such as *NACA 64-618* to reduce drag and enhance the lift-to-drag ratio, thereby maximizing energy capture. This blend of thick-to-thin airfoils aims to maintain efficient aerodynamic performance across the entire blade span, all while meeting the structural and geometric demands inherent in large-scale rotor systems.

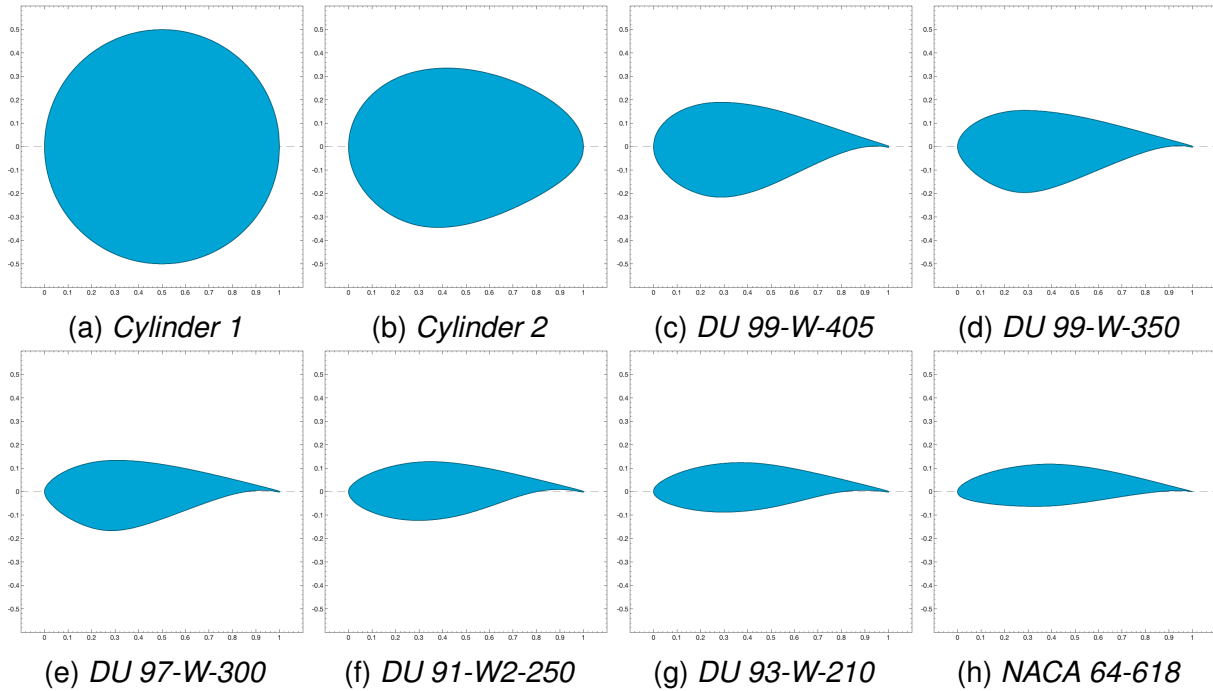


Figure 3.1: The eight airfoil profiles distributed along the blade span in [Figure 3.2](#)

Having established the new rotor diameter, the airfoil distribution is generated by straightforward similarity scaling from the original 5MW NREL reference blade. The same series of airfoil families—cylindrical root profiles transitioning into thick DU airfoils and ending in thinner NACA sections—are preserved to maintain overall aerodynamic characteristics. [Figure 3.2](#) illustrates a direct comparison of the airfoil layout for both the original reference blade and the scaled blade, highlighting how airfoil elements and their radial positions have been proportionally adjusted.

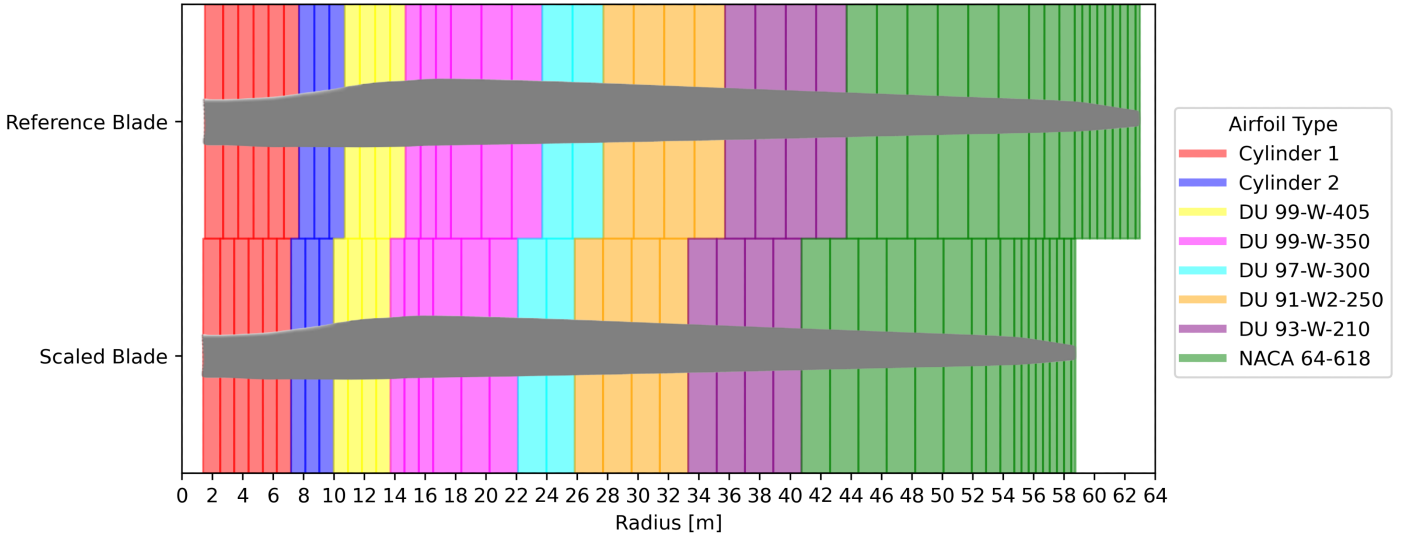


Figure 3.2: Comparison of the airfoil distribution between the original reference blade (63 m radius) and the scaled blade (58.75 m radius). Each color band indicates a distinct airfoil section.

3.2.2 Chord and Twist Distribution Design

The chord and twist distribution along the rotor blade, are established on scaling the reference turbine and empirical formulas describing the ideal distribution. Equation 3.1 and Equation 3.2 [25] are formulas based on the Blade Element theory with empirical corrections aiming to describe the ideal distributions.

$$\theta = \frac{\frac{2}{3}}{\lambda_{\text{design}} \left(\frac{r}{R} \right)} - \alpha_{\text{design}} \quad (3.1)$$

$$\frac{c}{R} = \frac{16}{9} \cdot \frac{\pi}{BC_l \lambda_{\text{design}}^2} \left(\frac{r}{R} \right)^{-1} \quad (3.2)$$

To establish the ideal blade performance, each airfoil's 2D aerodynamic properties are considered to provide the angle of attack α_{design} and the corresponding lift coefficient C_l . Consider the C_l/C_d and C_l/α plots for the NACA 64-618 airfoil shown in Figure 3.3 as an example. The optimal operation point corresponds to an angle of attack at where $(C_l/C_d)_{\text{max}}$. Even though choosing a higher angle of attack would provide a lift closer to the maximum lift coefficient $C_{l,\text{max}}$, limiting the angle of attack to $(C_l/C_d)_{\text{max}}$ creates a safety buffer to prevent the airfoil from stalling. If a wind turbine blade operating at $C_{l,\text{max}}$ experiences a wind gust, the lift coefficient becomes too high and starts to stall. This causes the turbulent separation point to occur earlier along the airfoil and introduces unwanted drag and unfavourable forces that could damage the structure of the blade. Repeating this approach for all the airfoils depicted in subsection 3.2.1 and using the resulting angle of attack's α and C_l in Equation 3.2 gives the analytical twist and chord distribution.

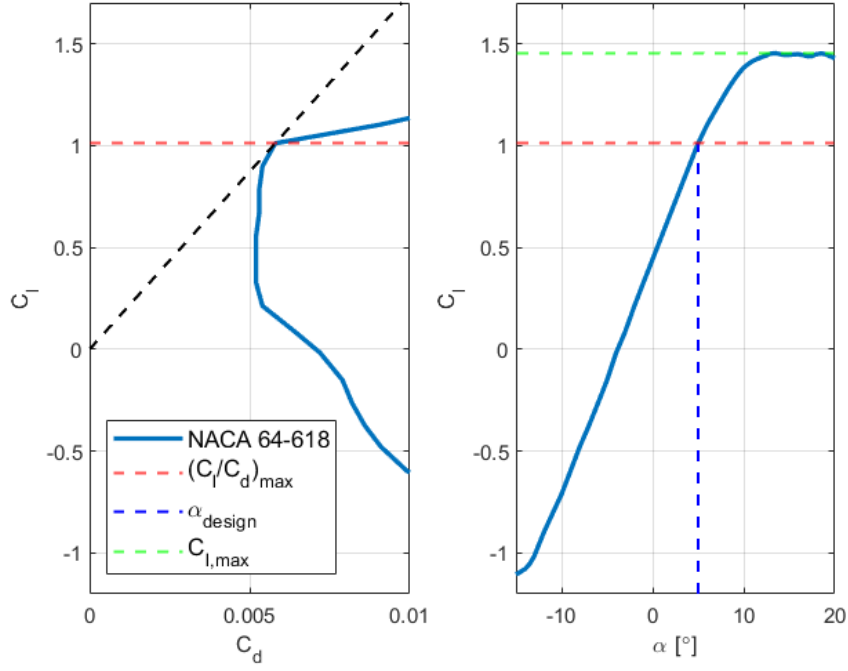


Figure 3.3: 2D airfoil characteristics illustrated in C_l/C_D and C_l/α plots.

The reference turbine chord and twists taken from [19] are scaled down to the appropriate blade length based on the linear scaling $c, \theta \propto R$. Figure 3.4 illustrates the scaled chord and twist distributions of the NREL reference turbine, along with the ideal airfoil performance described beforehand. A line of best fit is made for the ideal approach to showcase the behavior.

A noticeable difference between the two approaches is apparent near the root of the blade where the ideal approach predicts a significantly higher chord and twist angle. Even though these larger chords and twist values near the root are aerodynamically favorable, such large chords like $c = 10[\text{m}]$ or higher are structurally not feasible. Furthermore, considering that the majority of the energy is extracted near the tip of the blade, optimizing for the chord and twist near the root is not of significant importance. For these reasons, the chord and twist distribution for the turbine is optimized by taking a weighted average of the ideal approach's line of best fit and the scaled NREL reference for the range of $r/R = 0.35 - 0.9$ for the chord and $r/R = 0.35 - 1$ for the twist. The optimal chord distribution follows the chord distribution of the NREL reference turbine exactly near the tip $r/R > 0.9$ and near the root $r/R < 0.35$. An additional chord length of 0.5 [m] is added near the root to provide a smoother transition to the mid-span of the blade. The same approach is used for the twist distribution near the root, where it follows the scaled NREL distribution at $r/R < 0.35$.

The performance of the optimized distributions is analyzed by plotting the power coefficient C_P against the tip speed ratio λ for a range of pitch angles using the steady BEM algorithm. This curve is preferred over the power curve because it is directly related to the aerodynamic performance of the rotor instead of the entire system. Figure 3.5 shows that a pitch angle θ_p of 0° provides the highest C_P at a tip speed ratio of $\lambda = 7.6$.

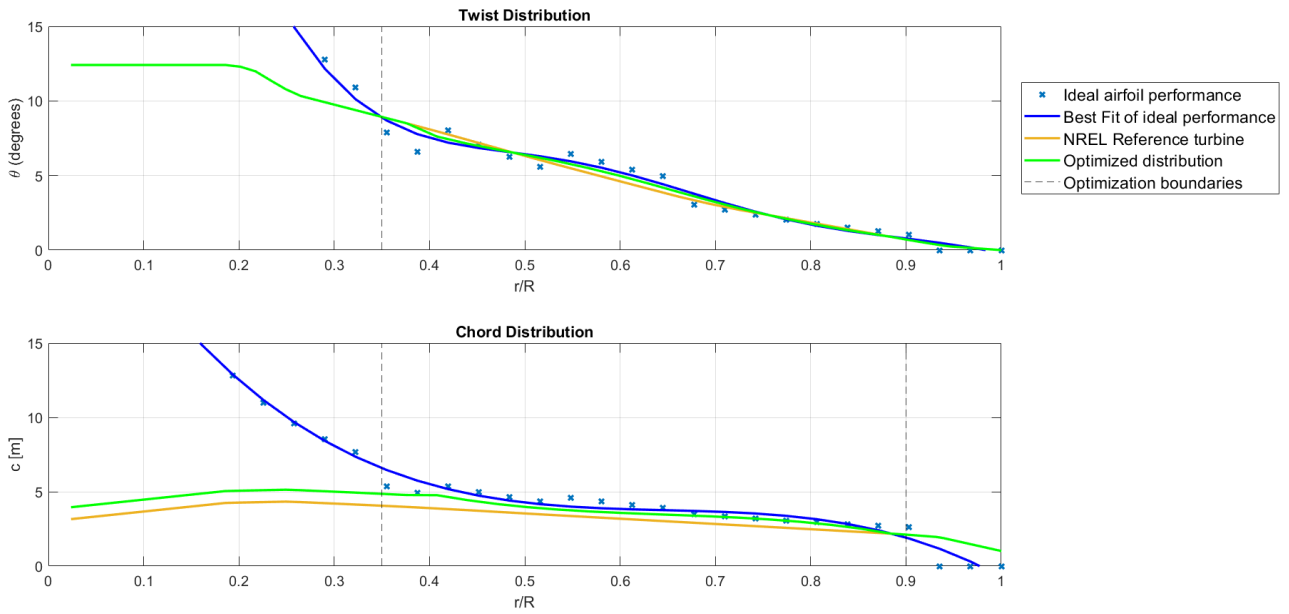


Figure 3.4: Scaled twist and chord distributions comparing the NREL reference turbine, ideal airfoil performance, fitted ideal curve, and optimized design.

Fine-tuning the chord for the ideal tip-speed ratio by implementing the linear scaling laws such that $c \propto \lambda^2$. When the twist is adjusted to keep the angle of attack the same, the new chord is given by Equation 3.3. Illustrating this visually in Figure 3.6, the fine-tuned chord is shown to exhibit a distribution similar to the NREL chord distribution.

$$\frac{c_{\text{new design}}}{c_{\text{initial design}}} = \frac{\lambda_{\text{initial design}}^2}{\lambda_{\text{new design}}^2} \quad (3.3)$$

With this new chord and twist distribution, the power coefficient is found to increase to a value of $C_P = 0.4802$ while keeping the tip speed ratio at $\lambda = 7.6$.

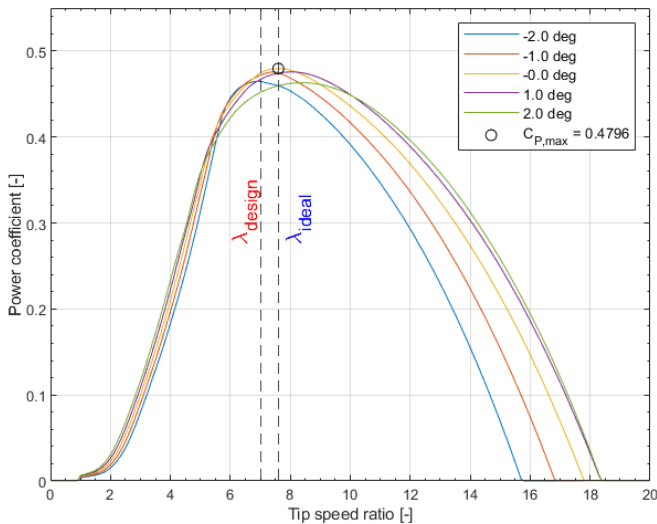


Figure 3.5: Steady-state power coefficient curves from FASTTool for varying blade pitch angles, showing sensitivity to tip speed ratio. The design and ideal operating points are marked.

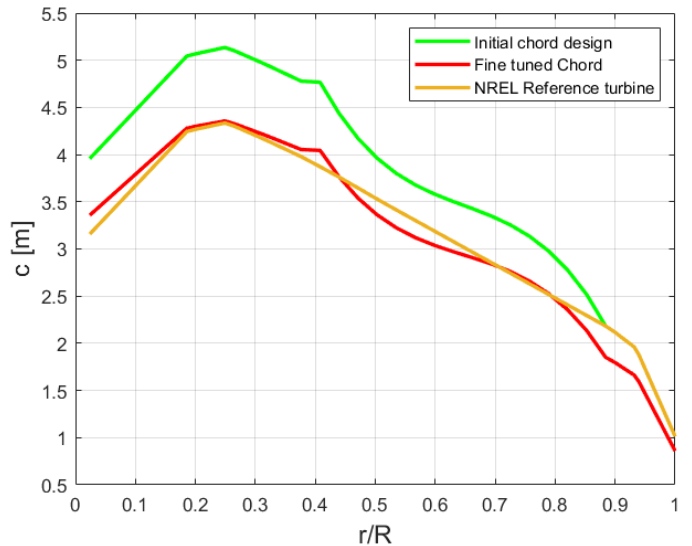


Figure 3.6: Comparison of initial, fine-tuned, and NREL reference chord distributions along the blade span.

3.2.3 Structural Properties and Scaling Laws

A well-established way to estimate the mass distribution of the blade is by using similarity scaling. In this approach, the mass per unit length,

$$m' = \rho A_{\text{cross-section}},$$

scales according to the square of the blade radius, R . Consequently, the total blade mass,

$$m = \int_0^R \rho A_{\text{cross-section}} dr,$$

scales according to R^3 . Based on the 5-MW reference turbine blade having a total mass of 17740 [kg] [19], the scaled blade mass is computed through

$$m \left(\frac{r}{R} \right)^3 = 17740 \text{ [kg]} \left(\frac{117.5[\text{m}]}{126[\text{m}]} \right)^3 = 14386.5 \text{ [kg]}.$$

Hence, the upscaled (or downscaled) blade mass follows directly from the cubic dependence on blade radius.

For the structural properties, the flapwise and edgewise bending stiffness, as well as the torsional stiffness, follow the $\propto R^4$ similarity law [11]. This implies that changes in blade length lead to a fourth-power scaling of the blade's overall bending and torsional rigidity.

Figure 3.7 shows the resulting mass distributions for both the reference blade and the newly scaled blade. Notably, the greatest portion of the mass is concentrated near the blade root—where the blade is structurally thickest and subjected to the largest bending moments. Similarly, Figure 3.8 presents the flapwise and edgewise stiffness distributions for the two blades, illustrating how the highest stiffness likewise resides in the blade root region. As expected, the stiffness distribution strongly correlates with the mass distribution, because more massive sections—built thicker for added strength—naturally achieve higher stiffness.

Table 3.3: *Key parameters comparing the reference blade and the scaled blade.*

Parameter	Reference Blade	Scaled Blade
Mass of blade [kg]	17740,0	14387,5
Rotor diameter [m]	126	117.5
Tip speed ratio [-]	7.55	7.6
Number of blades [-]	3	3
C_P [-]	0.482	0.4802

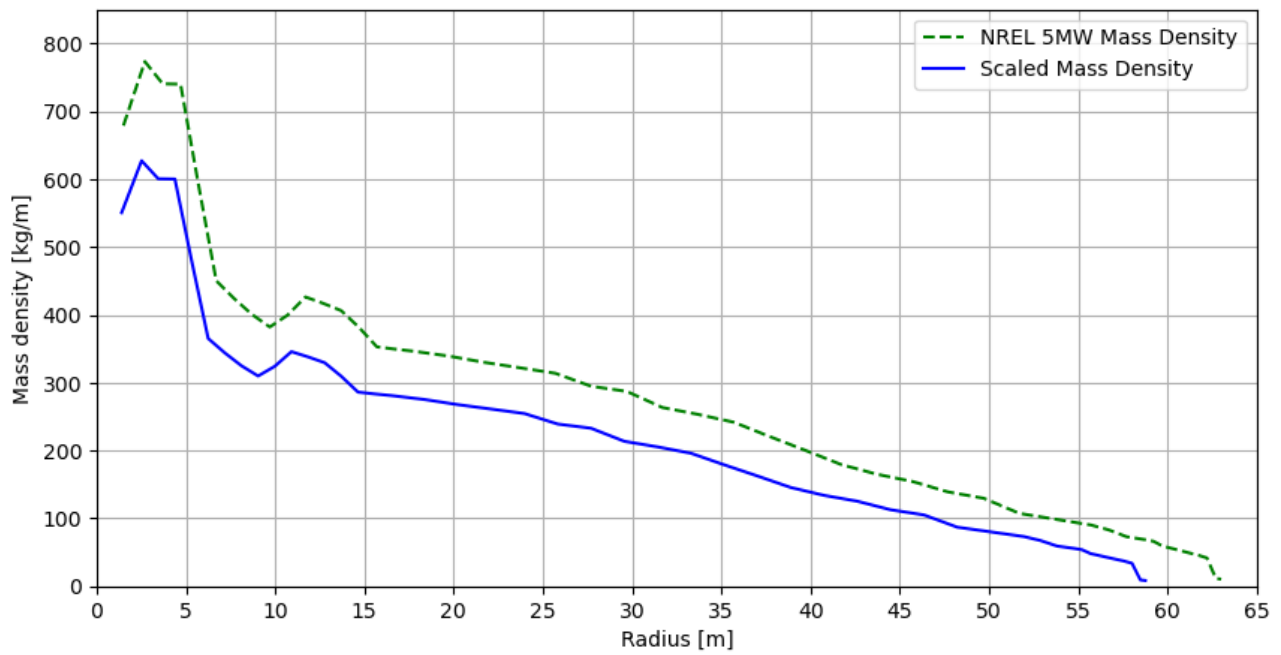


Figure 3.7: Comparison of the mass density distribution along the blade span, contrasting the original NREL 5MW blade (dashed) with the scaled blade (solid). The mass density is highest near the root, reflecting the structural reinforcement needed to carry the greatest bending loads.

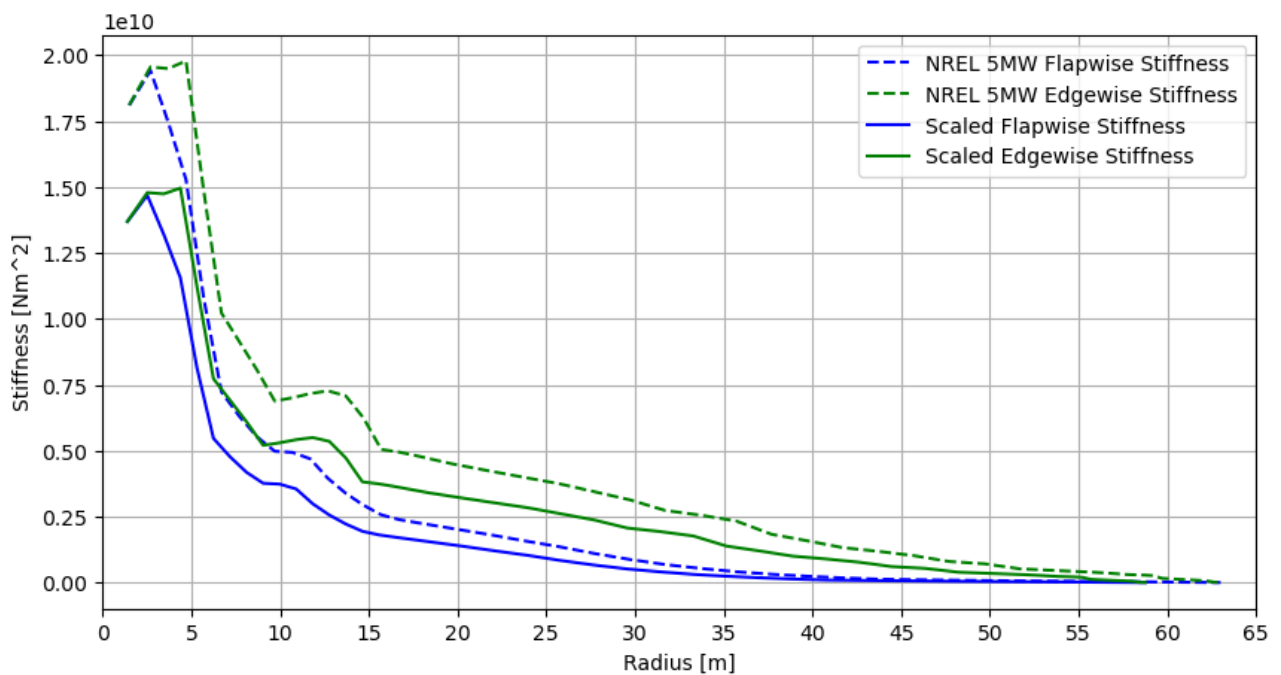


Figure 3.8: Flapwise and edgewise stiffness distributions for the NREL 5 MW reference blade (dashed) and the scaled blade (solid). Stiffness is greatest at the blade root, matching the region of highest mass concentration and structural demand.

3.2.4 Tip Deflection and Stresses

Tip deflection data for the NREL 5-MW reference turbine is obtained from the reference report [19], in which a maximum out-of-plane blade tip deflection of 5.5 [m] is reported. Geometric similarity scaling is applied to estimate the corresponding tip deflection for the scaled 3.5-MW turbine.

Assuming linear elastic behaviour and uniform material properties, the tip deflection can be expressed by a double integral over the bending moment distribution:

$$y_{\text{tip}} = \int_0^R \left(\int_0^r \frac{M(r')}{E \cdot I(r')} dr' \right) dr \Rightarrow y_{\text{tip}} \propto R \quad (3.4)$$

Thus, the maximum tip deflection scales proportionally with rotor radius, yielding the following estimate:

$$\max(y_{\text{tip}}) = 5.5 \text{ [m]} \cdot \frac{117.5 \text{ [m]}}{126 \text{ [m]}} = 5.13 \text{ [m]} \quad (3.5)$$

The original tip deflection curve was digitized from the reference plot, and the scaled deflection was then computed by applying the ratio of rotor radii. Figure 3.9 presents both the original and scaled tip deflections across a range of wind speeds.

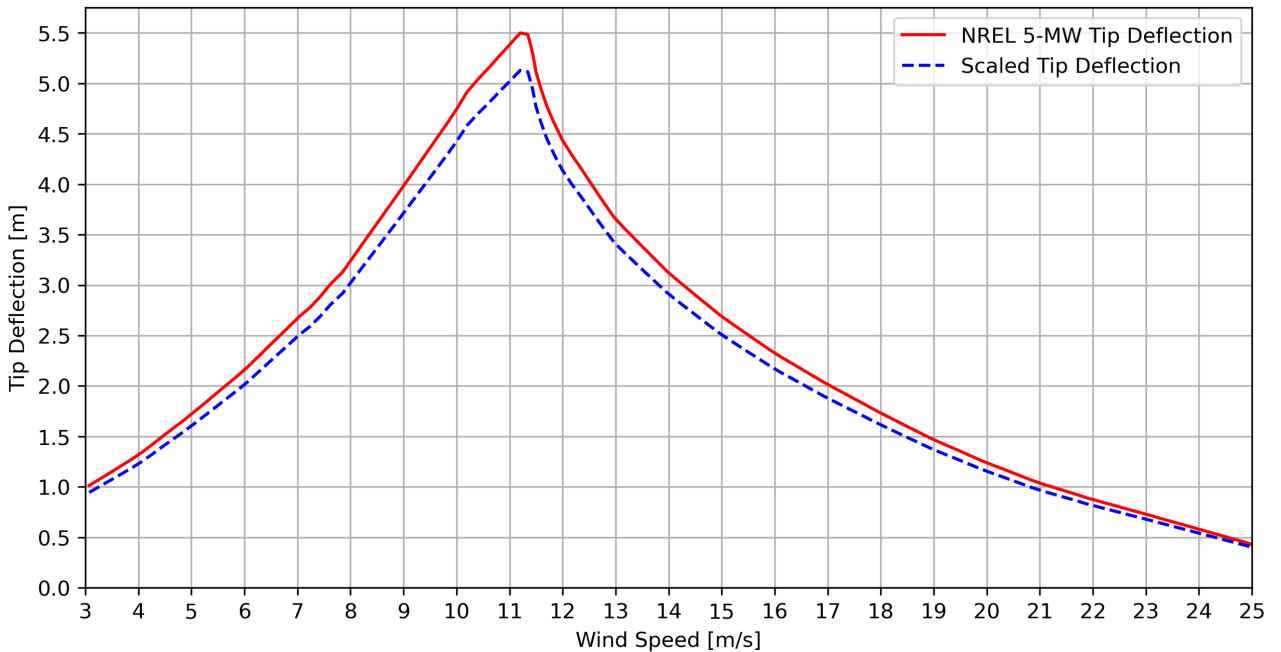


Figure 3.9: *Out-of-plane tip deflection for the NREL 5-MW reference turbine and the scaled tip deflection.*

The bending stress due to aerodynamic loading is independent of size based on linear geometric scaling laws $\sigma \propto R^0$. The only stress that scales with the radius is due to gravity. Considering that the bending moment imposed on the rotor blade due to gravity is given by $M_{\text{gravity}} = mg \cdot r$, the stress can be found using Equation 3.6.

$$\sigma_{\text{gravity}} = \frac{M_{\text{gravity}} y}{I_x} \quad (3.6)$$

The highest stress due to gravity is calculated to be 1.011×10^5 [N/m²]. Considering that NREL turbine is being downscaled while keeping the number of blades and the tip speed ratio roughly the same, the resulting stress and the tip deflection do not warrant changing the thickness factor from 1.0.

3.2.5 Overhang, Pre-bend, Tilt- and Cone-Angle

According to the NREL 5-MW reference report [19], the overhang distance is specified as 5 [m], the main shaft tilt angle is 5° , and the rotor cone angle is 2.5° upwind.

Using basic geometric considerations, this configuration yields a rotor clearance of more than 10.5 [m], which yields a safety factor of 1.9 when compared to the reported maximum out-of-plane tip deflection of 5.5 [m] [19].

Applying the same geometric principles to the scaled turbine with a rotor diameter of 117.5 [m] and an estimated maximum tip deflection of 5.13 [m], a safety factor of 1.5 can be obtained using a reduced main shaft tilt angle of only 3° . The resulting safety factor aligns with values typically deemed acceptable according to the literature [18]. This configuration results in a rotor clearance exceeding 7.6 [m]. Therefore, the main shaft tilt angle may be reduced to 3° while still retaining the original upwind cone angle of 2.5° and overhang of 5 [m].

Regarding blade pre-bending, the original NREL 5-MW turbine includes a built-in prebend to enhance tower clearance while minimizing rotor overhang. However, as noted in the reference report [19], the NREL 5-MW baseline turbine omits pre-bending and instead represents its effect through an upwind pre-cone angle of 2.5° . Following a similar approach, the present scaled design does not incorporate a built-in prebend in the blade geometry. Instead, a 2.5° upwind pre-cone angle is retained to emulate the structural clearance benefits typically provided by pre-bending while simplifying the blade modeling and ensuring compatibility with standard simulation tools and design practices.

Chapter 4

Drive Train Design

4.1 Drive train and electrical system configuration.

4.1.1 Summary of the design

Table 4.1: Drive train design variables/choices.

Parameter	Value / Description
Drive train configuration	Doubly-fed induction generator with gearbox (DFIG)

4.1.2 Supporting material, analyses and rationale

In order to evaluate the possible options, a trade-off is set up. Firstly, all possible options are presented and those not meeting the minimum requirements are instantly rejected in [subsubsection 4.1.2.1](#). Then the trade-off framework, including the determination of criteria, assignation of weights and a description score system, is described in [subsubsection 4.1.2.2](#). Finally, [subsubsection 4.1.2.3](#) performs the trade-off and establishes the winning configuration.

4.1.2.1 Description of options

Drive train configurations are always composed of a low-speed shaft that is directly or indirectly connected to the generator. The indirect connection involves a gearbox, which serves to increase the rotational speed of the shaft to match the required for the generator. The four most-used drive train options are the following:

- **Option A: Constant speed squirrel cage induction generator (CS)**

In the early years of wind turbines, this was the most used drive train type and was applicable to stall-controlled turbines. It consists of a three-stage gearbox that links the rotor with a squirrel cage generator, which is directly connected to the 50 Hz frequency

grid. Since the turbine is chosen to be a variable speed machine, this option is deemed to be unfeasible and, therefore, is not included in the trade-off below.

- **Option B: Doubly-fed induction generator with gearbox (DFIG)**

This drive train type consists of a multi-stage gearbox connected to the doubly-fed induction generator and frequency converter, allowing for stable grid connection. It is compatible for variable-speed machines and power reduction is achieved by pitching out the blades.

- **Option C: Brushless permanent magnet synchronous generator with gear and full converter (GFC)**

In order to obtain superior grid-fault ride-through characteristics than DFIG [23], the DFIG block is exchanged for a brushless, permanent magnet, synchronous generator. This drive train configuration is compatible with variable-speed machines and, hence, considered a feasible option in the trade-off.

- **Option D: Direct drive generator**

To limit the maintenance due to the failures in the gearbox, this component is taken out of the drive train, resulting in the generator being directly connected to the rotor. These low-speed generators are rather large and consist of significantly more pole pairs compared to other two types presented (Option B and Option C). Similarly to other options, it is fully compatible with variable-speed machine.

4.1.2.2 Trade-off framework

The trade-off framework includes a description of the criteria used, assigning weight to each criterion and defining the score scale. Criteria are determined based on system requirements, the operational point of view and differences between the feasible options. With the established criteria, their weights are assigned ensuring that all add up to 1 and reflect criteria significance in relation to others. The criteria, with the corresponding explanation of weights, is given below:

- **Efficiency** - The power extracted from the machine is the most important design parameter. As the rotor efficiency is limited by aerodynamics, any losses in the drive train, whether electrical or mechanical, should be minimized. Following this train of thought, the criterion is deemed the most important and awarded a **weight of 0.25**.
- **Complexity and Maintenance** - Complex multi-component systems tend to have lower system reliability, as there are more points of failure. Moreover, these failures increase the maintenance services required and the associated costs. Given its high importance, this criterion is awarded **the baseline weight of 0.2**.
- **Power Quality** - Power quality encompasses mainly two aspects of the electrical de-

sign of the drive train. The first is whether the system allows for voltage and frequency control, as well as how much the so-called "flicker" is an evident problem. The second aspect is how well the system is dealing with grid faults. Given its high importance, this criterion is awarded **the baseline weight of 0.2**.

- **Cost** - Every engineering decision must be justified economically, meaning that at after a given point saving up money is chosen over enhanced performance. As a result, the criterion is considered to be important and awarded **the baseline weight of 0.2**.
- **Mass** - The mass of the drive train system is one of the factors driving tower design, as well as the size and structure of the nacelle. Therefore, the objective is always to minimize mass while maximizing performance. Nevertheless, this criterion does not contribute directly to the drive train performance. Therefore, given its lower importance, it has been awarded **the lower weight of 0.15**.

Having established the criteria and their corresponding weights, the scoring scale shall be defined. It is decided that all options are going to be assessed using a 1-5 scale. [Table 4.2](#) provides an overview of each score definition.

Table 4.2: *Score definition for drive train trade-off*

Score	Meaning
1	Unacceptable performance
2	Correctable defeciences
3	Sufficient performance
4	Good performance
5	Excellent performance

4.1.2.3 Trade-off

The trade-off results are presented in [Table 4.3](#) and [Table 4.4](#). Each cell presents a score based on its colour, as defined in [Table 4.2](#), and features a small explanation as to why it received such score. The overall score of each design option is given in the last column of [Table 4.4](#), calculated based on the weights of the different trade-off criteria. Note that the table has been split in two for ease of reading.

Table 4.3: *Trade-off comparison - part 1*

	Complexity & Maintenance	Efficiency	Cost
Weights	0.2	0.25	0.2
Direct Drive	No gearbox removes source of unreliability and thus decreases maintenance frequency.	Higher efficiency due to elimination of gearbox losses, specially compared to multi-stage gearbox systems [25].	Higher initial capital costs due to large generator size, specially for larger turbines which may present challenges to manufacture.
DFIG	Slip rings are a major source of reliability issues, requiring frequent maintenance and repair of slip rings and brushes [13].	Lower overall efficiency due to high gearbox power losses, although this is partially offset by reduced generator and converter losses [25].	Lower initial capital cost due to smaller generator and partial power converter, although gearbox costs also should be considered.
GFC	Much higher converter failure rates compared to DFIG, requiring much more frequent maintenance and replacement [10].	Permanent magnets display higher efficiency, especially when operating well below full power (i.e. at part load) [13].	Permanent magnet generators are typically more expensive and face greater price volatility risks due to availability of rare earth metals [13].

Table 4.4: *Trade-off comparison - part 2*

	Mass	Power Quality	Total
Weights	0.15	0.2	1
Direct Drive	Higher generator diameter due to lower rotational speed of the shaft, resulting in higher mass [13].	Full converter system allows for full control over grid currents, reducing flicker and harmonics while providing good voltage and frequency support [24].	3.30
DFIG	Smaller converter required, potentially reducing tower loading if it is placed inside the nacelle. [13].	System can provide reactive power control, but it is susceptible to grid disturbances (e.g. voltage dips) which can affect stability [24].	3.35
GFC	Higher generator mass due to integration of additional power electronics and cooling systems [25].	Similar to direct drive, the full converter allows for precise control of power output, reducing flicker and harmonics while ensuring stable voltage and frequency regulation [24].	3.25

To summarise, the Direct Drive system scores well in efficiency and power quality due to the elimination of gearbox losses and the use of a full converter. However, its higher initial cost

and increased generator mass pose challenges. The GFC system provides high efficiency and power quality similar to Direct Drive, but suffers from higher maintenance due to converter failure rates and higher generator costs associated with permanent magnet materials.

DFIG, on the other hand, presents a balanced performance across all criteria. Although it suffers from moderate power quality concerns due to grid disturbances, it benefits from a lower initial cost and reduced overall system mass, making it the most favourable option with the highest overall score of 3.35. Therefore, moving forward, the DFIG-based drive train is selected as the most suitable configuration for the 3.5MW onshore wind turbine design, as presented in [Table 4.1](#).

4.2 Generator

4.2.1 Summary of the design

Table 4.5: *Generator objectives and boundary conditions*

Parameter	Value / Description
Power rating	3.5 MW
Maximum Rotational Speed (HSS)	1250 rpm
Maximum Torque (HSS)	26.55 kNm

Table 4.6: *Generator design variables/choices.*

Parameter	Value / Description
Type of generator	Doubly-fed Induction Generator (DFIG)
Number of poles of the generator	6
Voltage level	690 V

4.2.2 Supporting material, analyses and rationale

To determine the number of pole pairs of the generator, [Equation 4.1](#) can be used [22]. f represents the grid frequency, which for Europe is 50 Hz, and n_s is the synchronous speed of the generator. To determine this value, a reference speed of 1200 rpm from a similar generator with comparable characteristics is used as a starting point [7]:

$$p = \frac{60f}{n_s} = \frac{60 \cdot 50}{1200} = 2.5 \approx 3 \quad (4.1)$$

Note that the number of pole pairs is rounded up to meet the minimum requirements, as it is not physically possible to have 2.5 pole pairs. The output voltage level of the generator is set at 690 V, as is standard for generators of this type.

4.3 Gearbox

4.3.1 Summary of the design

Table 4.7: *Gearbox objectives and boundary conditions*

Parameter	Value / description
Maximum Rotational Speed (LSS / HSS)	12.76 / 1250 rpm
Maximum Torque (LSS / HSS)	2602 / 26.55 kNm
Minimum gearbox ratio	97:1

Table 4.8: *Gearbox design choices for gearbox*

Parameter	Value / description
Number of stages	3
Type of stage 1 & 2	Planetary
Transmission ratio stage 1 & 2	7:1
Type of stage 3	Parallel
Transmission ratio stage 3	2:1
Total gear ratio	98:1

4.3.2 Supporting material, analyses and rationale

To determine the required gearbox ratio for the chosen 6-pole, Doubly Fed Induction Generator, the following approach can be taken, as per [22]:

- **Determine the Speed Range:** The speed range of the generator depends on the converter rating, which has an estimated value of $x \approx 0.25$. Assuming a synchronous speed of $n_s = 1000$ rpm, as found in [22], the minimum and maximum shaft rotational speeds are calculated as:

$$n_{\min} = (1 - x \cdot \sqrt{3})n_s = 567 \text{ rpm} \quad (4.2)$$

$$n_{\max} = (1 + x)n_s = 1250 \text{ rpm} \quad (4.3)$$

- **Calculate the Gearbox Ratio:** The gearbox ratio GR is the ratio of the maximum generator speed $n_{\max}$ and the maximum rotational speed of the rotor $n_{r_{\max}} = 12.85$ rpm, as calculated in Table 2.2 :

$$GR = \frac{n_{\max}}{n_{r_{\max}}} = \frac{1250}{12.85} \approx 97 \quad (4.4)$$

In order to achieve the required gearbox ratio of 97, a choice was made to have a three-stage gearbox. This consists of two planetary stages, each with a gear ratio of 7:1, followed

by a parallel-shaft gear stage, with a gear ratio of 2:1. This combination coincides with those found in literature for turbines of similar power ratings. In total, this gives the gearbox a gear ratio of 98:1, meeting the minimum requirement of 97:1.

4.4 Converter

4.4.1 Summary of the design

Table 4.9: *Converter objectives and boundary conditions*

Parameter	Value / Description
Voltage level	690 V
Minimum Power rating	3.5 MW
Grid Frequency	50 Hz

Table 4.10: *Converter design variables/choices*

Parameter	Value / Description
Location	Tower Base
Type of converter	ACS880-87LC-4000A/3800A-7 (Back-to-back configuration)

4.4.2 Supporting material, analyses and rationale

Before choosing converter type, its location is determined. Considering that converter is a relatively heavy component, which preferably should be easily accessible by electrical technicians, it is logical to position it at the tower base. Nevertheless, such a positioning comes at the disadvantage of higher losses in the connecting cable from the generator. To minimize those losses, the converter could be positioned in the nacelle, which would increase its mass and consequently influence the tower design. Therefore, to minimize nacelle mass, the decision is made to place the converter at the tower base.

For the turbine at hand, an off-the-shelf converter is chosen. A back-to-back converter ACS880-87LC-4000A/3800A-7 is chosen, as it complies with the voltage and current requirements [6]. Moreover, it is able to match the European grid frequency.

4.5 Power Cables in Tower

4.5.1 Summary of the design

Table 4.11: *Power cable objectives and boundary conditions*

Parameter	Value / Description
Voltage level	690 V
Current rating per phase	2790 A

Table 4.12: *Power cable design variables/choices*

Parameter	Value / Description
Type of cables	EVA LSZH Torsion Resistant 1.8/3kV Turbine Cable
Length	90 m

4.5.2 Supporting material, analyses and rationale

Due to the yaw system in the tower, the cable linking the generator and the converter has to be torsion-resistant. Moreover, it has to be able to sustain voltage and current ratings as specified in Table 4.11, as derived from the generator output power and voltage level. The length of cable is approximated to be 90 meters, sufficient to go from the nacelle to the tower base. To comply with all these requirements, the EVA LSZH Torsion Resistant 1.8/3kV Turbine Cable is chosen [9]. Note that this cable is designed to handle higher voltage levels than that of the current design, however it also features a higher cross-sectional area. This will help reduce cable power losses at the expense of a higher cable cost, but this was deemed to be justifiable to improve the efficiency of the turbine.

4.6 Nacelle and Drivetrain Properties

4.6.1 Summary of the design

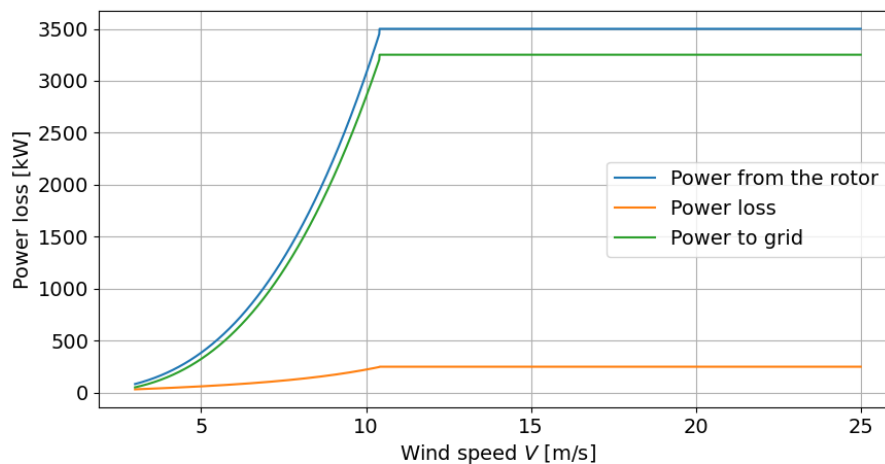


Figure 4.1: *Power output from the rotor, power delivered to grid and system power loss as a function of wind speed.*

Table 4.13: Drive train properties

Parameter	Value / Description
Turbine efficiency	0.92
Gearbox efficiency	0.97
Generator efficiency	0.98
Converter efficiency	0.97
Cable efficiency	0.99
Nacelle mass [kg]	125,981
Nacelle cost [\$]	1,143,253
Generator inertia [kgm ²]	3,843

4.6.2 Supporting material, analyses and rationale

In this subsection, the reasoning and intermediate steps are explained in the process of nacelle component sizing. Firstly, mass and cost estimation of drivetrain components are outlined in [subsubsection 4.6.2.1](#). Next, dimensions of primary drivetrain components are determined in [subsubsection 4.6.2.2](#). Finally, power losses per component are calculated in [subsubsection 4.6.2.3](#).

4.6.2.1 Mass and cost estimation of drive train components

In this subsection, the mass of different components in the nacelle are evaluated. The estimations are done based on the NREL report [20]. The results with their associated equations are presented in [Table 4.14](#). The total nacelle mass is calculated to be approximately 126 tonnes. This value is validated against the nacelle of the Siemens Gamesa 3.6-107 wind turbine, which has a mass of 125 tonnes [17]. Additionally, alongside with mass, the cost of each component and total nacelle assembly is estimated.

Table 4.14: Mass and cost estimation of nacelle-mounted components based on [20]

Needed parameters				
Rotor Diameter D [m]		117.5		
Rotor Radius R [m]		58.75		
Low-speed shaft torque Q [kNm]		2,770		
Machine rating [kW]		3,500		

Component	Mass		Cost	
	Equation	Value [kg]	Equation	Value [\$]
Blade	$m_{blade} = 0.4948 \cdot R^{2.53}$	14,791.75	-	-
Hub	$m_{hub} = 0.954 \cdot m_{blade} + 5680.3$	19,791.63	$C_{hub} = m_{hub} \cdot 4.25$	84,114
Total pitch bearing system	$m_{pitch} = 0.1295 \cdot 3 \cdot m_{blade} + 5680.3$	6,237.91	$C_{pitch} = 1.328 \cdot m_{pitch} + 555$	8,839
Nose cone	$m_{nose} = 18.5 \cdot D - 520.5$	1,653.25	$C_{nose} = m_{nose} \cdot 5.57$	9,209
Low-speed shaft	$m_{shaft} = 0.0142 \cdot D^{2.888}$	13,506.96	$C_{shaft} = 0.01 \cdot D^{2.887}$	9,467
Main bearings	$m_{bearing} = (D \cdot 8/600 - 0.033) \cdot 0.0092 \cdot D^{2.5}$	2,111.61	$C_{bearing} = 2 \cdot m_{bearing} \cdot 17.6$	74,329
Gearbox	$m_{gearbox} = 70.94 \cdot T^{0.759}$	29,089.30	$C_{gearbox} = 16.45 \cdot \text{machine rating}^{0.759}$	439,245
Brake/coupling system	$m_{brake} = C_{brake}/10$	696.29	$C_{brake} = 1.9894 \cdot \text{machine rating} - 0.1141$	6,963
Generator	$m_{generator} = 6.47 \cdot \text{machine rating}^{0.9223}$	12,011.53	$C_{generator} = \text{machine rating} \cdot 65$	227,500
Yaw system	$m_{yaw} = 1.6 \cdot 0.0009 \cdot D^{3.314}$	10,434.41	$C_{yaw} = 2 \cdot 0.0339 \cdot D^{2.964}$	92,645
Mainframe	$m_{mainframe} = 2.233 \cdot D^{1.953}$	25,745.34	$C_{mainframe} = 9.489 \cdot D^{1.953}$	104,714
Hydraulic&cooling system	$m_{hydraulic} = 0.08 \cdot \text{machine rating}$	280	$C_{hydraulic} = \text{machine rating} \cdot 12$	42,000
Nacelle cover	$m_{cover} = C_{cover}/10$	4,422.92	$C_{cover} = 11.537 \cdot \text{machine rating} + 3849.7$	44,229
Total:		125,981	Total:	1,143,253

Despite the fact that converter is not placed in the nacelle, its mass is 3500 kg as specified in product specifications [6] and reported here for completeness.

4.6.2.2 Dimensions of drivetrain components

Dimensions of each component are estimated using simple engineering models. Firstly, the chosen DFIG generator is modeled as a solid cylinder. Based on the currently existing generators for similar power turbines, it is estimated to have a stator radius of $r_s = 0.80$ m and a stack length of $l_s = 3.50$ m [5]. Overall dimensions of the generator including the casing are given in Table 4.15. With the determined stator radius, an estimate for generator inertia can be made according to the following formula:

$$I_{generator} = \frac{1}{2} m_{generator} r_s^2 \quad (4.5)$$

Substituting in the values for stator radius and mass of the generator (from Table 4.14), the inertia is computed to be 3,843 kgm².

Secondly, the dimensions on the gearbox are estimated by referencing to the existing one for the similar turbine rating described in [15]. When it comes to dimensions of the chosen off-the-shelf converter and power cables, they are specified in the product specifications in [6] and [9] respectively. Note that three cables are needed, as it is a three-phase system. Dimensions of all the components are reported in Table 4.15.

Table 4.15: *Dimensions and total volume of drivetrain components*

Component	Length (m)	Height (m)	Depth (m)	Total Volume (m ³)
Generator [5]	3.50	2.25	1.70	13.39
Gearbox [15]	3.20	2.28	2.05	14.96
Converter [6]	2.50	2.00	1.20	6.00
Cable [9]	90	0.004	0.004	$3 \times 0.036 = 0.108$

4.6.2.3 Power losses of drivetrain

To estimate losses of each component of the drivetrain, the methodology outlined by Henk Polinder in [25] is followed. Gearbox and converter losses are simply modeled using their respective efficiencies as described by Polinder [25], whereas generator is modeled using values and equations found in the report [25]. When it comes to cable losses calculations, they can be determined by computing the current flowing out of the converter and evaluating losses due to cable material resistivity. The results of the calculations are presented as a function of wind speed in Figure 4.2. It can be noticed that all components except the gearbox follow power curve profile, as expected based on the results achieved by Polinder [25].

Having determined power losses per component, it is possible to calculate total power loss of the system P_{loss} and, consequently, power delivered to the grid P_{grid} according to the equation:

$$P_{grid} = P_{rotor} - P_{loss}, \quad (4.6)$$

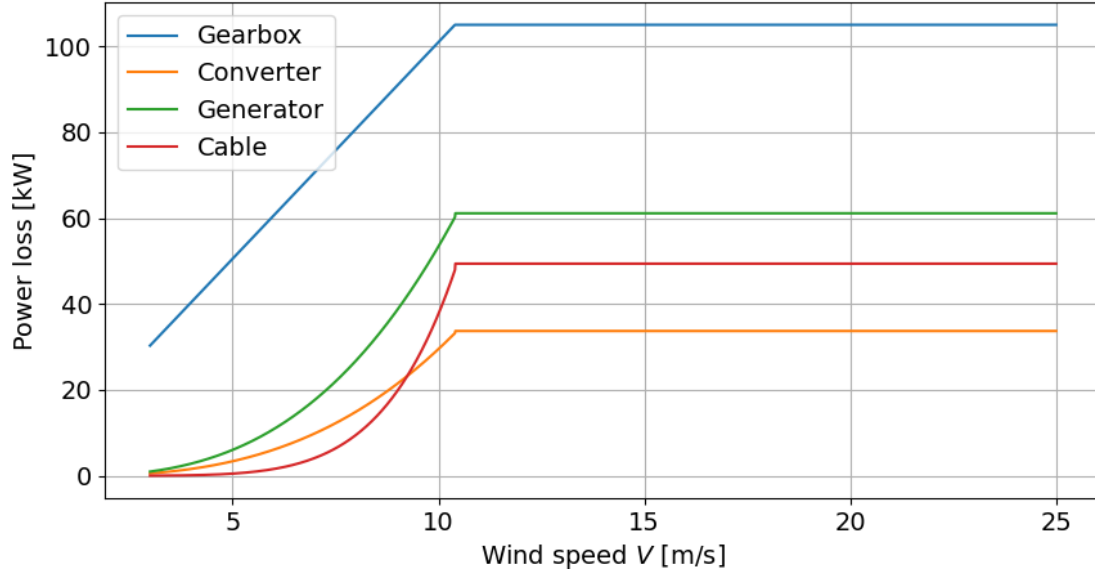


Figure 4.2: *Power losses per component as a function of wind speed.*

where P_{rotor} stands for power produced by the rotor. The results of this computations are presented in Figure 4.1. Additionally, the efficiency of the system, which is defined as the ratio between power delivered to grid P_{grid} to the mechanical power from the rotor P_{rotor} , is plotted in Figure 4.3. The plot shows an increase in efficiency up until rated wind speed, where a maximum efficiency of around 92% is achieved. Note that, in the following analysis, the transformer is not included and its losses are not modeled as it is out of the scope of this project. Nevertheless, transformers are highly efficient machines and losses from that component are relatively negligible compared to others.

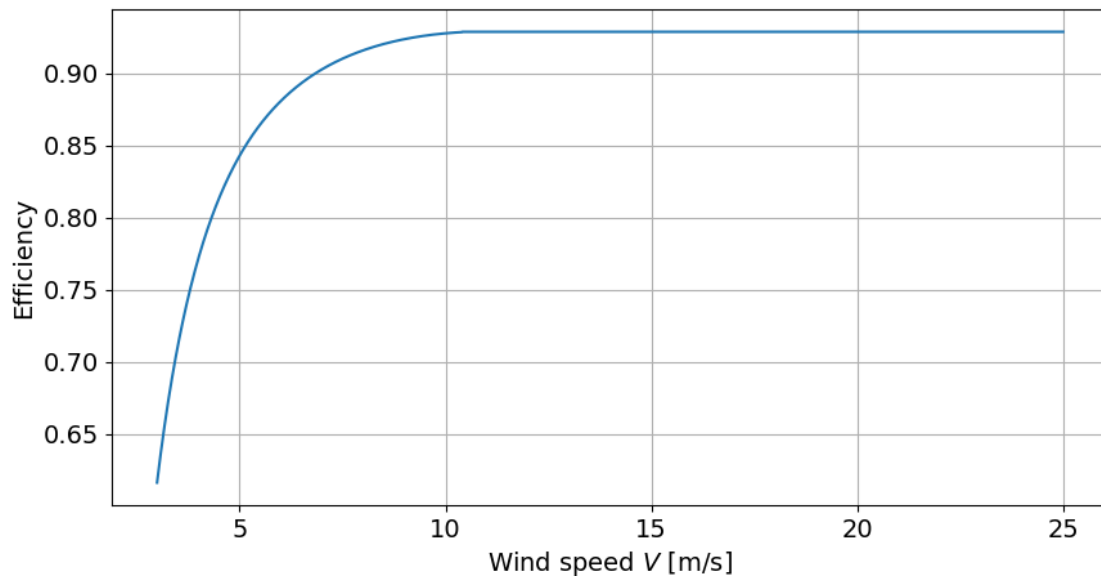


Figure 4.3: *Efficiency of the drivetrain system.*

Chapter 5

Steady State and Dynamics

5.1 Summary of the outcome and updates

Table 5.1: *Objectives and boundary conditions tower design*

Parameter	Value / description
Ratio between hub heights of new turbine and reference turbine	84 m / 90 m (0.9333)
Ratio between (maximum) thrusts of new turbine and reference turbine	552 kN / 721 kN (0.7656)
Optimal fine pitch angle	0 degrees
Optimal tip speed ratio	7.6

Table 5.2: *Design variables / choices tower design*

Parameter	Value / description
Tower top diameter	3.61 m
Tower top wall thickness	0.0233 m
Tower base diameter	5.36 m
Tower base wall thickness	0.0313 m

Table 5.3: *Design variables / choices control curve*

Parameter	Value / description
Control generator speed A	352.286 rpm
Control generator speed B	457.972 rpm
Control generator speed B2	1209.023 rpm
Control generator speed C	1221.236 rpm
Optimal mode gain	1.715 Nm/rpm ²
Demanded torque	28790.03 Nm

Table 5.4: *Changed objectives and boundary conditions*

Parameter	Value / description
Ratio between hub heights of new turbine and reference turbine	84 m / 90 m (0.9333)
Ratio between (maximum) thrusts of new turbine and reference turbine	552 kN / 721 kN (0.7656)
Optimal fine pitch angle	0 degrees
Optimal tip speed ratio	7.6

Table 5.5: *Changed design variables / choices*

Parameter	Value / description
Tower base diameter	6.968 m
Tower base wall thickness	0.0407 m
Control generator speed A	567 rpm
Control generator speed B	737.1 rpm
Control generator speed B2	1209.023 rpm
Control generator speed C	1221.236 rpm

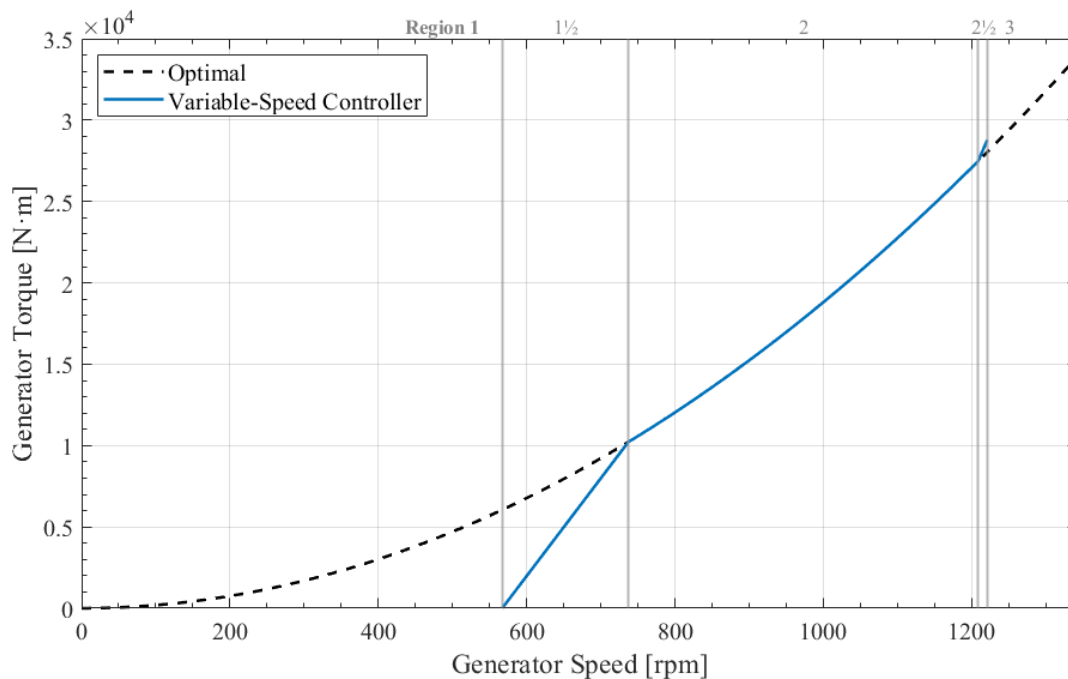


Figure 5.1: *Torque control curve of the updated design*

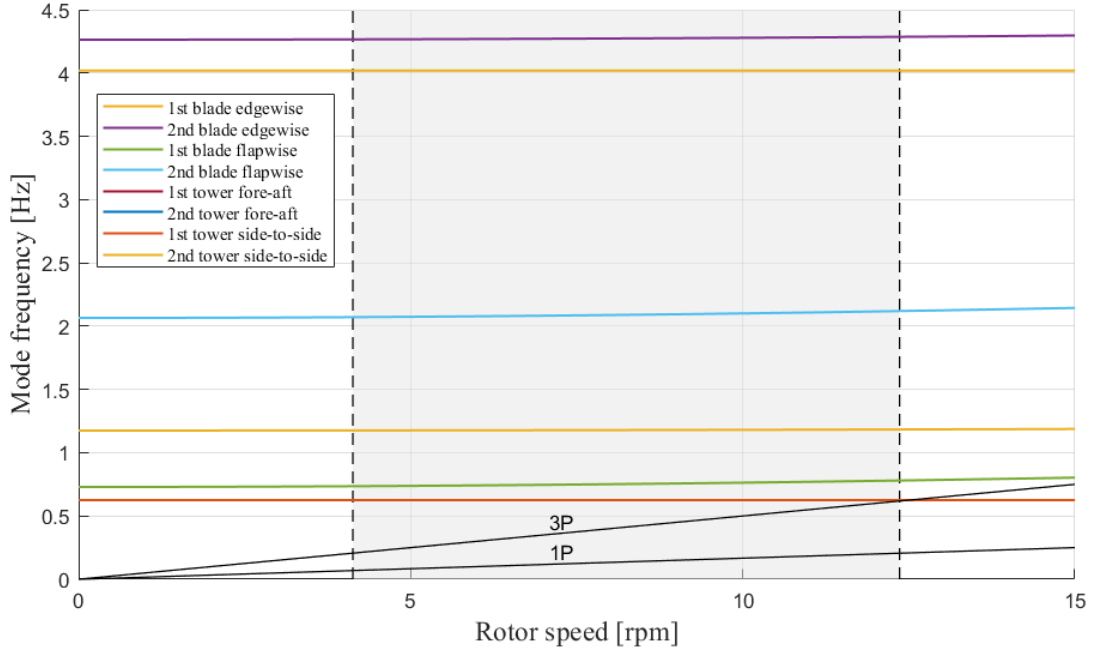


Figure 5.2: *Campbell diagram of the updated design*

5.2 Supporting material, analyses and rationale

5.2.1 Torque control

This section aims to provide the rationale behind the selection and tuning of the design variables for the control curve as listed in [Table 5.3](#) and [Table 5.5](#).

5.2.1.1 Demanded torque

To determine the demanded torque of the generator, [Equation 5.1](#) is used with P_g being the generator power of 3.5MW, ω_r the rated rotor speed of 12.847 [rpm], and GR the gearbox ratio of 98 as determined in [section 4.3](#).

$$T_g = P_g / \omega_g = GR \cdot P_g / \omega_r, \quad (5.1)$$

After applying the gearbox and generator efficiencies found in [section 4.6](#) (the converter efficiency has been incorporated into the generator, and cable losses are neglected), a value of 28790.03 [Nm] is obtained. Although the torque limit is not required for now, an estimate can be calculated by applying the same margin of 10%, as in the reference turbine [\[19\]](#), which leads to a torque limit of 31669.33 [Nm].

5.2.1.2 Control generator speeds

The torque control curve shown in [Figure 5.1](#) is characterised by the following speed set-points that allow for effective transitions across operational regions. The first one to be determined is generator speed C , which marks the start of Region 3 and the full load regime. It therefore corresponds to the rated rotational speed of the high speed shaft, which is nothing else than the rated speed of the turbine times the gear ratio and gear efficiency as seen below in [Equation 5.2](#), which yields 1221.236 [rpm].

$$\omega_C = \omega_r \cdot GR \cdot \eta_{gearbox} \quad (5.2)$$

The next speed to be determined is the control generator speed B2, which is the transitional generator speed between Regions 2½ and 3 and can be taken as 99% of the generator speed C [19], resulting in 1209.023 [rpm]. It is worth noting that the reference documentation is inconsistent with this number, as it sometimes uses 98% instead.

The control generator speed A marks the end of Region 1 and represents the minimum shaft speed. This can be calculated by using the same process used in [Equation 5.2](#) but using the minimum rotor speed as seen in [Equation 5.3](#) instead of the rated [19]. However, taking the minimum rotor speed of the turbine at the design tip speed ratio (3.706 [rpm]) would result in a minimum shaft speed of 352.286 [rpm], which is below the minimum rotational shaft speed of 567 [rpm] found in [section 4.3](#). Therefore, the control generator speed A is taken to be 567 [rpm]. This entails having a higher tip speed ratio during this operating region, with its other effects being discussed in greater detail below in [subsection 5.2.2](#).

$$\omega_A = \omega_{rmin} \cdot GR \cdot \eta_{gearbox} \quad (5.3)$$

The final control generator speed left to be determined is generator speed B , which indicates the start of Region 2 and can be taken as an increment of 30% from the control generator speed A , or 737.1 [rpm].

5.2.1.3 Optimal mode gain

The final input parameter that needs to be calculated is the optimal mode gain, which can be easily found by applying [Equation 5.4](#) with the values determined in [subsection 3.2.2](#) and [section 4.3](#), resulting in 1.715 [Nm/rpm²].

$$K_{opt} = \frac{1}{2} \rho \pi R^5 \frac{C_p}{\lambda^3} \frac{1}{(GR \cdot \eta_{gearbox})^3} \quad (5.4)$$

The remaining parameters for torque control, namely the torque minimum, the torque slew rate, and the generator speed low-pass cut-off, shall be kept unchanged.

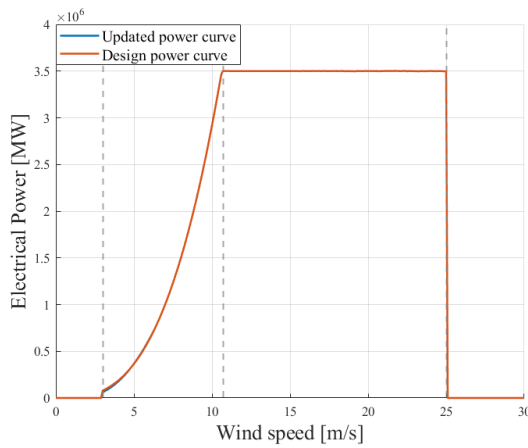
5.2.2 Performance analysis

The actual performance of the turbine differs from the design objectives established in the previous chapters due to the efficiency losses found in Chapter 4, these variations have been compiled into [Table 5.6](#).

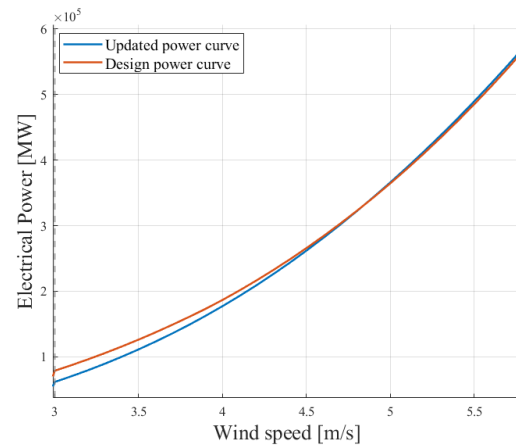
Table 5.6: Comparison of design and obtained values for key wind turbine parameters

Parameter	Design Value	Obtained Value
λ	7.6	7.2767
C_p	0.4802	0.4763
V_{rated} [m/s]	10.4	10.7
ω_{rated} [rpm]	12.847	12.4594

It is important to note that the optimal tip speed ratio and power coefficient designed in Chapter 3 are not reached during partial load. Both the rated rotor speed and the rated wind speed decrease according to the 97% efficiency of the gearbox. This directly affects the tip speed ratio and consequently the power coefficient. However, these losses are too minimal to justify any changes to the design. The performance of the turbine can be best assessed by computing the power curve, which is a system-dependent graph showing the variation of power with wind speed. The effects of raising the control generator speed A to 567 [rpm] on the power produced by the turbine can be seen in [Figure 5.3](#). The overlap between the



(a) Full power curves for both generator speed control strategies



(b) Enlargement of section with differing power generation

Figure 5.3: Comparison of power curves for both generator speed control strategies.

curves from 5 [m/s] onward is an indicator that both systems have entered Region II and operate at the shared optimal conditions. It is apparent that the control strategy with a lower control generator speed A reaches this region quicker, as it also has a lower generator speed B. However, the losses in power production are not substantial and although these occur at wind speeds that according to the Weibull curve from Chapter 2 are expected to be frequent, the overall decrease on annual energy production can be considered acceptable.

5.2.3 Tower design

The tower design procedure consists of first deriving a scaling law for the tower base that keeps the stress due to the bending moment constant and then applying linear scaling with the rotor diameter to obtain the tower top dimensions, with in-between dimensions being linearly interpolated. As part of this, the ratio between the tower base diameter and tower base wall thickness is required to remain constant.

To calculate the bending moment experienced by the tower, only the thrust of the rotor is considered. Neglecting the effects of gravity and drag, because of the assumption that they are significantly smaller than the thrust. The bending moment is therefore taken at the tower base as the rotor thrust force times the hub height, as expressed in [Equation 5.5](#).

$$M = F_{thrust} \cdot h_{hub} \quad (5.5)$$

The thrust can be obtained from the FASTTool Matlab workspace, which shows a maximum thrust force of 552540.18 [N]. The stress due to the bending moment is expressed in [Equation 5.6](#), where r is half of the tower base diameter and I is the moment of inertia about the centroidal axis for a hollow cylinder as defined in [Equation 5.7](#).

$$\sigma = \frac{Mr}{I} \quad (5.6)$$

$$I = \frac{\pi}{4}(r_{outer}^4 - r_{inner}^4) = \frac{\pi}{64}(D_{outer}^4 - D_{inner}^4) \quad (5.7)$$

Since the thickness of the tower wall is much smaller than the diameter ($D/t = 0.0058$), the approximation in [Equation 5.8](#) may be used for the moment of inertia.

$$I \approx \frac{\pi D^3 t}{8} \quad (5.8)$$

Substituting into [Equation 5.6](#), the expression in [Equation 5.9](#) is obtained.

$$\sigma \approx \frac{8M}{\pi D^2 t} \quad (5.9)$$

Therefore, to keep the stress constant, it follows that $D^2 t \propto M$. Furthermore, since the ratio between tower base diameter and tower base wall thickness is kept constant, a first-order scaling approach to avoid buckling is used such that $D/t = \text{constant}$ and giving $D^3 \propto M$ or $D \propto M^{1/3}$. Consider the following.

$$D_{b,scaled} = D_{b,original} \left(\frac{F_{scaled} \cdot h_{scaled}}{F_{original} \cdot h_{original}} \right)^{1/3} = 6 \left(\frac{5.52 \times 10^5 \cdot 84}{7.21 \times 10^5 \cdot 90} \right)^{1/3} = 5.36 \text{ [m]} \quad (5.10)$$

$$t_{b,scaled} = t_{b,original} \left(\frac{F_{scaled} \cdot h_{scaled}}{F_{original} \cdot h_{original}} \right)^{1/3} = 0.035 \cdot \left(\frac{5.52 \times 10^5 \cdot 84}{7.21 \times 10^5 \cdot 90} \right)^{1/3} = 0.0313 \text{ [m]} \quad (5.11)$$

The dimensions of the tower top have been found by applying the ratio between rotor diameters as seen below in Equation 5.12 and Equation 5.13:

$$D_{t,scaled} = D_{t,original} \left(\frac{r}{R} \right) = 3.87 \cdot \left(\frac{117.5}{126} \right) = 3.61 \text{ [m]} \quad (5.12)$$

$$t_{t,scaled} = t_{t,original} \left(\frac{r}{R} \right) = 0.025 \cdot \left(\frac{117.5}{126} \right) = 0.0233 \text{ [m]} \quad (5.13)$$

The interpolated values between the base and top are shown in Table 5.7. The last tower height is input 2.5 [m] (half of the nacelle height) below the hub height as per FASTTool guidelines.

Table 5.7: *Dimensions of the scaled tower*

Height [m]	Diameter [m]	Wall thickness [m]
0.000	5.3600	0.0313
8.150	5.1850	0.0305
16.300	5.0100	0.0297
24.450	4.8350	0.0289
32.600	4.6600	0.0281
40.750	4.4850	0.0273
48.900	4.3100	0.0265
57.050	4.1350	0.0257
65.200	3.9600	0.0249
73.350	3.7850	0.0241
81.500	3.6100	0.0233

Table 5.8: *Modal analysis frequencies comparison of reference, scaled, and updated turbine designs (Hz)*

Mode	Reference		Scaled		Updated	
	1st	2nd	1st	2nd	1st	2nd
Tower fore-aft	0.34	3.07	0.42	3.28	0.62	4.02
Tower side-side	0.34	3.07	0.42	3.28	0.62	4.02
Blade flapwise (0 rpm)	0.68	1.93	0.73	2.06	0.73	2.06
Blade edgewise (0 rpm)	1.10	3.97	1.18	4.26	1.18	4.26

Using the 'Modal Analysis' function, the natural frequencies of the tower and rotor are displayed in Table 5.8, while the natural frequencies for the reference turbine are displayed in Table 5.8. Comparing their respective Campbell diagrams in Figure 5.4 and Figure 5.5 clearly shows how the scaled turbine presents resonance risks for the 3rd period as it crosses the tower fore-aft and side-to-side frequencies at about 8.45 [rpm]. The reference turbine avoids these risks through a soft-stiff design, as the 0.34 [Hz] natural frequencies of the tower remain within the 0.20-0.345 [Hz] range imposed by the 1st and 3rd periodic excitations. There is a risk of resonance with the 3rd period at the start of the operating range at

about 6.9 [rpm], but since the loads are smaller at that speed, the risk can be managed with an appropriate control strategy.

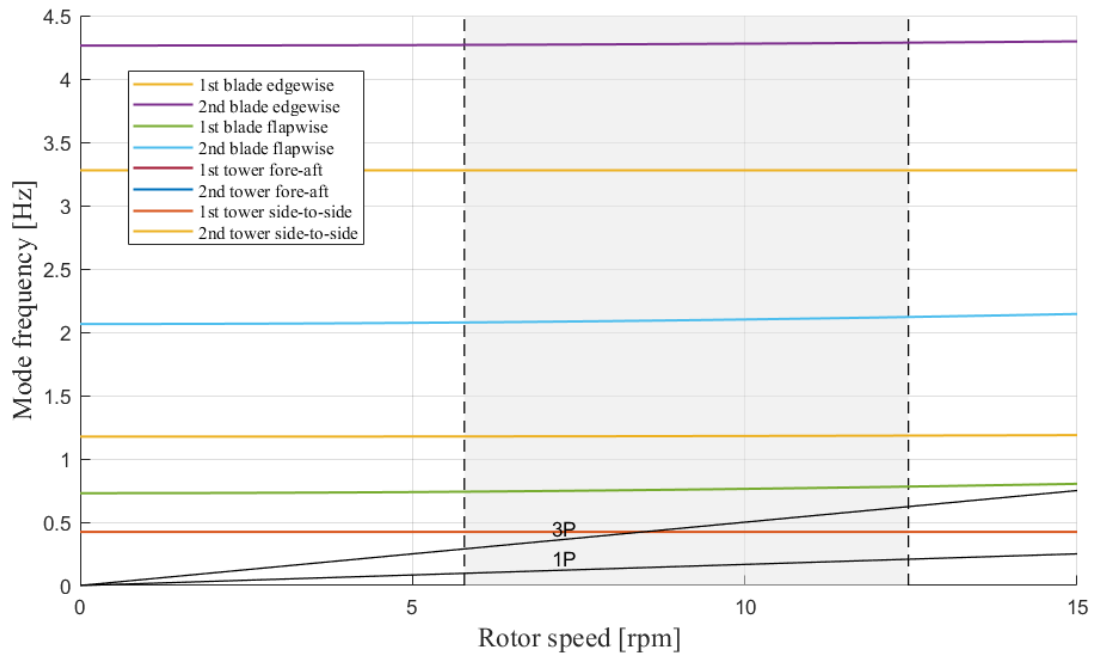


Figure 5.4: *Campbell diagram of the scaled turbine*

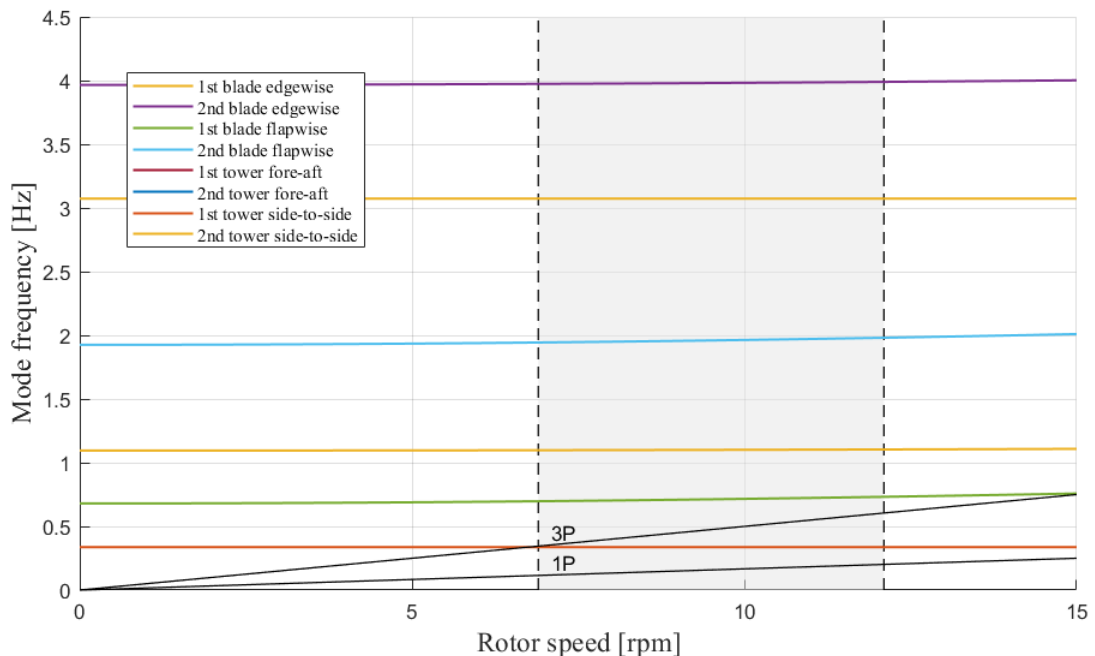


Figure 5.5: *Campbell diagram of the reference turbine*

To address the severe risk of resonance in our turbine, the design must be modified to shift the natural frequencies of the tower. Besides curtailing the operational speed range, modifying the dimensions of the tower is a viable alternative. While a soft-stiff approach might

be preferable for its reduced material costs, the window offered by the 1st and 3rd periods is rather narrow (0.20-0.29 [Hz]) without considering safety margins. A stiff-stiff solution is therefore chosen, which would require the natural frequencies of the tower to be above the 0.623 [Hz] as imposed by the upper bound of the 3rd period. As seen in Equation 5.14, the tower base diameter is the parameter with the largest effect on the stiffness, and is therefore increased by 30%. The thickness was also increased to ensure that the ratio remained constant as per the design requirements.

$$f_n \propto \sqrt{\frac{k}{m}} = \sqrt{\frac{EI}{mL^3}} = \sqrt{\frac{E\pi D^3t}{8mL^3}} \quad (5.14)$$

With this configuration, the new properties of the tower are as shown in Table 5.9 and Table 5.8, and its Campbell diagram is displayed in Figure 5.2. Observing closely, the tower frequencies are now of 0.625 [Hz], narrowly avoiding crossing with the third harmonic line.

Table 5.9: *Dimensions of the updated tower*

Height [m]	Diameter [m]	Wall thickness [m]
0.0000	6.9680	0.0407
8.1500	6.6322	0.0390
16.3000	6.2964	0.0372
24.4500	5.9606	0.0355
32.6000	5.6248	0.0337
40.7500	5.2890	0.0320
48.9000	4.9532	0.0303
57.0500	4.6174	0.0286
65.2000	4.2816	0.0268
73.3500	3.9458	0.0250
81.5000	3.6100	0.0233

The updated design avoids severe resonance risks, but there is a remaining mild risk of resonance at the upper end of the operating range, when the turbine is under full load conditions with wind speeds exceeding 10.7 [m/s]. An effective control strategy and implementing safety margins is needed to reduce fatigue loading under turbulent conditions or gusts, as the dynamic tower responses will require special attention under these conditions. Regarding blade tip deflection, there are no immediate indicators that suggest it will exceed design limits, but this should be verified later, as increased tower top deflection might affect tip clearance.

Chapter 6

Control Design

6.1 Description of the control system

Wind turbine control can be divided into operational control, which determines how control objectives are met between cut-in and cut-out wind speeds, and load control, which aims to mitigate the resonance effect due to approaching first tower eigenfrequency, i.e. a tower fore-aft damping controller. The operational control is carried out by two decoupled controllers, pitch and torque controllers, as it is depicted in Figure 6.1.

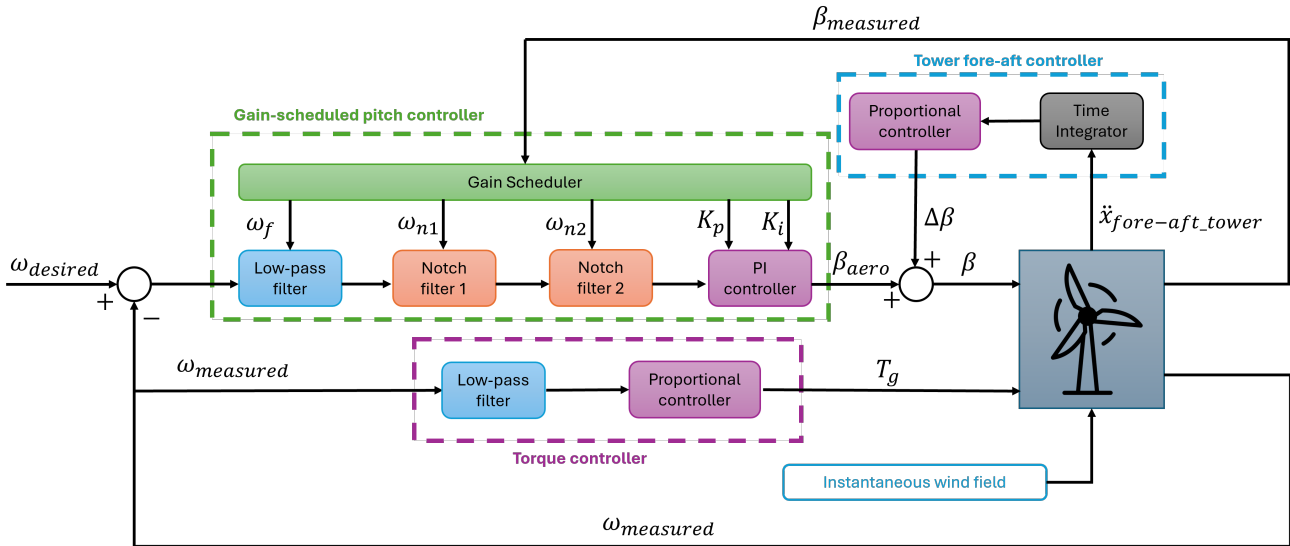


Figure 6.1: Block diagram of the design wind turbine control system including operational pitch and torque controllers, as well as load control. The wind turbine block represents non-linear dynamics including sensors and actuators.

Each controller operates within a defined range of wind speeds. Namely, the torque controller is activated below the rated windspeed (the so-called Region 2) to change the turbine rotational speed to ensure maximum power extraction from the flow. By adjusting the generator gain, the optimal tip speed ratio for power output is tracked. On the other hand, the pitch

controller changes the pitch angle of the blades above the rated wind speed (the so-called Region 3). The primary objective is to limit aerodynamic loads and to keep the power level at the value of the rated power. The characteristics of Region 2 and Region 3 are summarised in Table 6.1.

Table 6.1: *Control region characteristics. Control strategy below cut-in and above cut-out speeds is beyond the scope of this analysis.*

Characteristic	Region 2	Region 3
Wind Speeds	Between cut-in and rated wind speed	Between rated and cut-out wind speed
Controller	Torque Controller	Pitch Controller
Actuator	Power Converter	Pitch Motor
Aim	Track optimal tip speed ratio Maximize power extraction	Limit aerodynamic loads Extract constant rated power
Rotational Speed	Varying	Fixed
Blade Pitch Angle	Fixed	Varying

The transition between Region 2 and Region 3 causes the two controllers to interfere with each other. Hence, a smooth activation of the pitch controller and deactivation of the torque controller is required. Piecewise linear control, where the system is linearised in different regions, each with its own controller, is used in this report. Other controller switch methods are beyond the scope of this report. However, the possible solution is to define an additional control region (Region 2.5) with gradual attenuation of the torque controller action and activation of the pitch controller. Unfortunately, a discontinuous slope in the torque control curve can contribute to the excessive loading on the turbine, which is undesirable [21].

The architecture of the torque controller, as depicted in Figure 6.1, consists of a low-pass filter to attenuate high frequencies and a subsequent proportional gain controller. On the other hand, the pitch controller architecture is more sophisticated. The control is exercised using gain scheduling, which means that the gains and filter frequencies used change depending on the state of the system; the measured blade pitch angle. The signal of the error in the rotational speed is passed through a low-pass filter to attenuate high-frequency content and notch filters to tackle interfering resonance frequencies. Lastly, the scheduled gains are applied to the signal, resulting in resulting collective blade pitch angle β_{aero} that is given to the actuation mechanism.

The tower fore-aft damping controller is a load controller which aims to mitigate the fore-aft oscillations of the tower, its architecture is shown in Figure 6.1. The tower fore-aft acceleration is measured and integrated, resulting in a computed tower velocity. The velocity value is multiplied by a proportional gain, resulting in an additional change in the pitch angle required $\Delta\beta$. This change is added to the pitch angle β_{aero} stemming from the pitch control, giving the resulting collective pitch angle β passed to the actuator.

6.2 Controller tuning

Pitch controller tuning is accomplished by loop-shaping in the frequency domain. The linearised nominal loop system model is combined with a controller transfer function, resulting in the desired performance. The controller is tuned to satisfy the requirements relating to gain margins ($GM > 6$ [dB]), phase margins ($PM > 40$ [deg]) and gain-crossover frequencies ($\omega_{gc} \approx 0.2\pi$ [rad/s]). The controller gains are initially tuned at pitch angles being 2.73, 13.52, 19.02 and 24.81 degrees, as pictured in Figure 6.2, while the gains for the rest are interpolated.

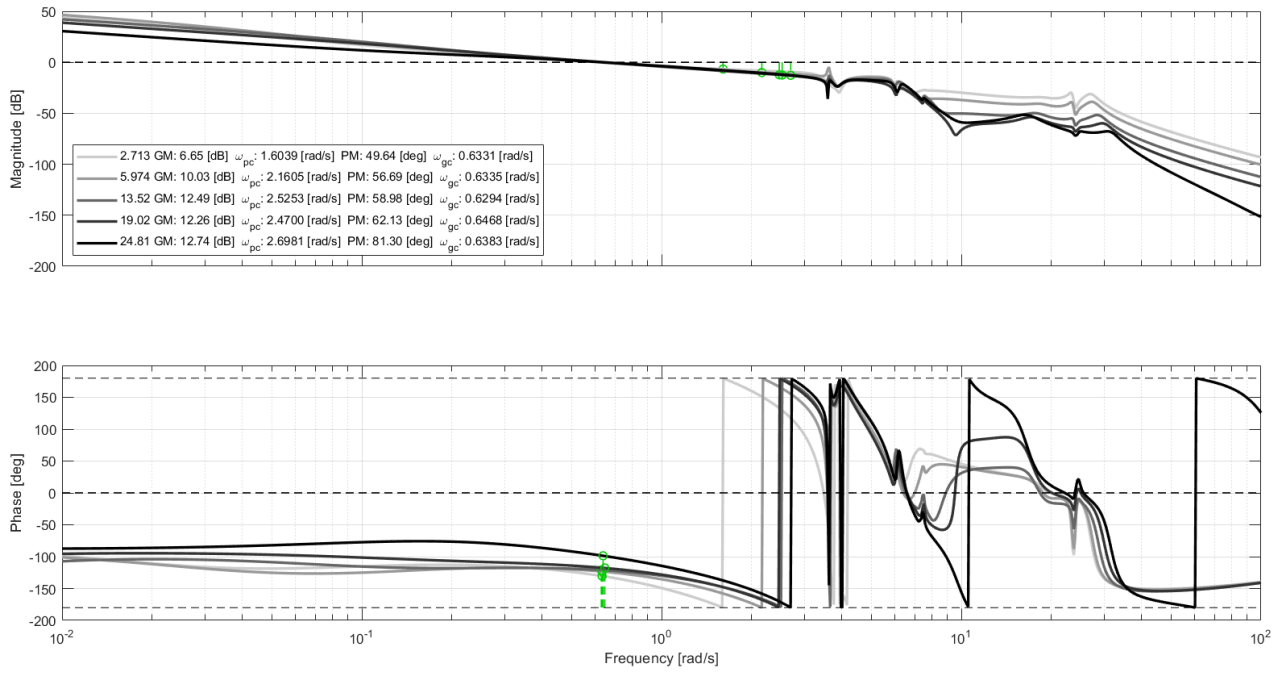


Figure 6.2: Bode plots of loop-shaped controller for the selected pitch angles.

As the gains for pitch angle equal to 2.73 degrees are substantially higher than for other angles, the pitch angle of 5.974 degrees is made separately, and those gain values are used for interpolation. The gain-scheduling parameters are defined in Table 6.2, where β and γ are constants defining first and second notch filters, respectively.

Table 6.2: Gain-scheduling for the pitch controller.

Angle β [deg]	K_p	K_i	β_1	β_2	ω_{n1}	γ_1	γ_2	ω_{n2}	ω_{filter}
2.713	-0.045	-0.0030	0.01	0.1	3.61	0.01	0.1	5.95	3
5.974	-0.025	-0.0029	0.01	0.1	6.06	0	0	0	3
8.271	-0.019	-0.0028	0.01	0.1	6.06	0	0	0	3
10.20	-0.016	-0.0027	0.01	0.1	6.06	0	0	0	3
11.93	-0.014	-0.0026	0.01	0.1	6.06	0	0	0	3
13.52	-0.012	-0.0025	0.01	0.1	6.06	0.105	0.055	7.875	3
15.00	-0.0112	-0.0024	0.01	0.12	6.06	0	0	0	3
16.39	-0.0103	-0.0023	0.01	0.14	6.06	0	0	0	3
17.73	-0.0095	-0.0022	0.01	0.16	6.06	0	0	0	3
19.02	-0.0086	-0.0021	0.01	0.18	6.06	0	0	0	3
20.25	-0.0078	-0.0020	0.01	0.2	6.06	0	0	0	3
21.45	-0.0076	-0.0018	0.01	0.175	6.075	0	0	0	3
22.60	-0.0073	-0.0015	0.01	0.15	6.09	0	0	0	3
23.72	-0.0071	-0.0013	0.01	0.125	6.105	0	0	0	3
24.81	-0.0068	-0.0010	0.01	0.1	6.12	0	0	0	3

To tune the tower fore and aft damping controller, several values of damping coefficient are tested in time-domain turbulent simulations. Time-domain response is converted using FFT to the frequency domain, presented in [Figure 6.3](#), where it is analysed. As the purpose of the controller is to decrease the tower acceleration close to the first natural frequency ($\omega = 0.62$), the value of the damping gain $K = 0.18$ is selected as it corresponds to better signal attenuation with smaller highest peak in that region than the ones pertaining to gains $K = 0.15$ and $K = 0.2$. Moreover, it should be noted that for the damping gain $K = 0.3$, the signal is amplified, which translates to the case of negative damping and, consequently, some kind of system instability.

6.3 Performance evaluation of the controller

The pitch controller is evaluated in time domain simulations by qualitative analysis of step response and normal turbulence model simulations with time-varying wind velocity. Firstly, the difference between the gain-scheduled controller and the fixed gain one. The gains for the fixed gain controller are determined as the average of the gains given in [Table 6.2](#). As a consequence of it, the cross-over frequency requirement is no longer satisfied as it varies significantly, whereas the gain and phase margins requirements are met. From the step response of both controllers, presented in [Figure 6.4](#), it can be deduced that the fixed gain controller is characterised by higher overshoot and rise time (especially for lower wind speeds) compared to the gain-scheduled controller. Moreover, for higher wind speeds, higher vibrations of the tower in side-to-side motions are observed for the gain-scheduled controller. Similar

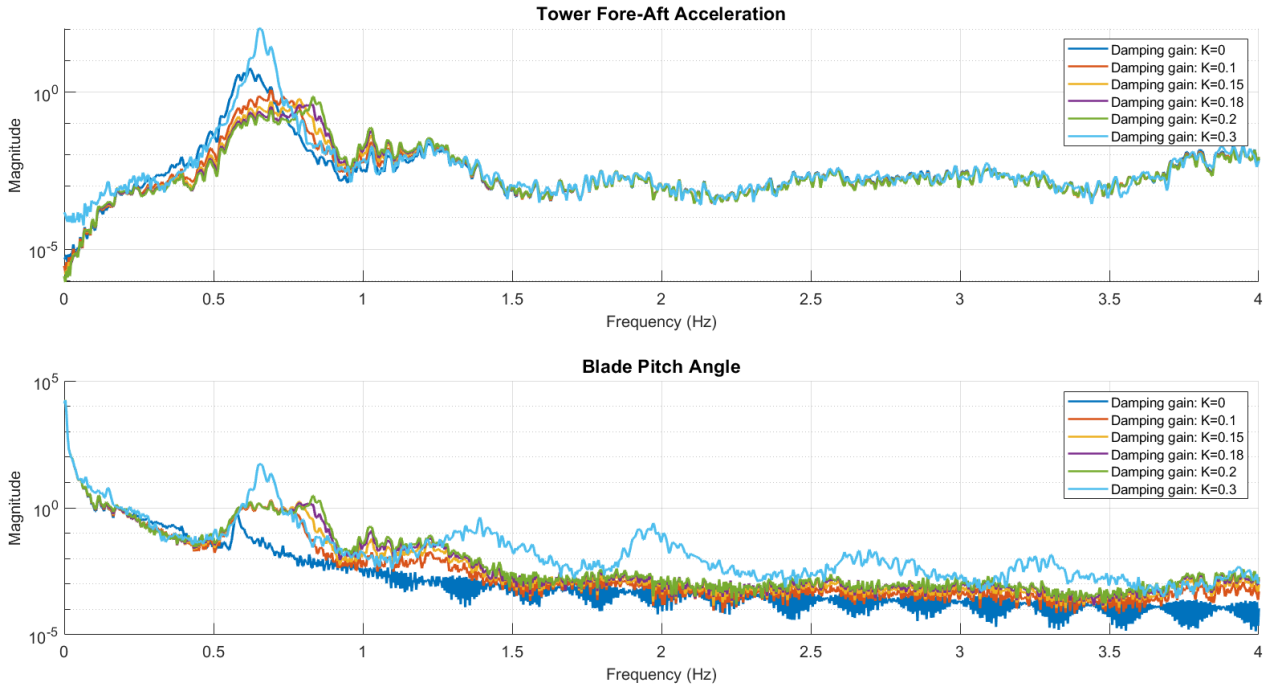


Figure 6.3: *Tower fore-aft acceleration and blade pitch angle signals in the frequency domain for low frequencies. The signal is obtained by running a normal turbulence model (NMT) simulation for a duration of 660s with the average wind speed of 15 m/s, and the spectrum is smoothed out using `pwelch` function in MATLAB®.*

conclusions are drawn from a system response to turbulent inflow conditions, shown in [Figure 6.5](#). Therefore, the gain-scheduled controller proves to performs better in comparison to the fixed gain one.

The effect of the tower fore-aft damping controller is assessed similarly by time-domain simulations of the step response, as well as turbulent inflow conditions. From the step response of the system, presented in [Figure 6.6](#), no visible difference between the damped and undamped responses can be seen. On the other hand, the turbulent inflow conditions simulation response, illustrated in [Figure 6.7](#), shows that the damping controller significantly reduces the magnitude of the tower fore-aft acceleration. Consequently, it also helps to attenuate the side-to-side tower oscillations, but alas, to a lesser extent. Moreover, the frequency domain analysis of the fore-aft acceleration signal shown in [Figure 6.3](#) allows us to conclude that there is a significant reduction of the acceleration magnitude in the region of the first tower natural frequency.

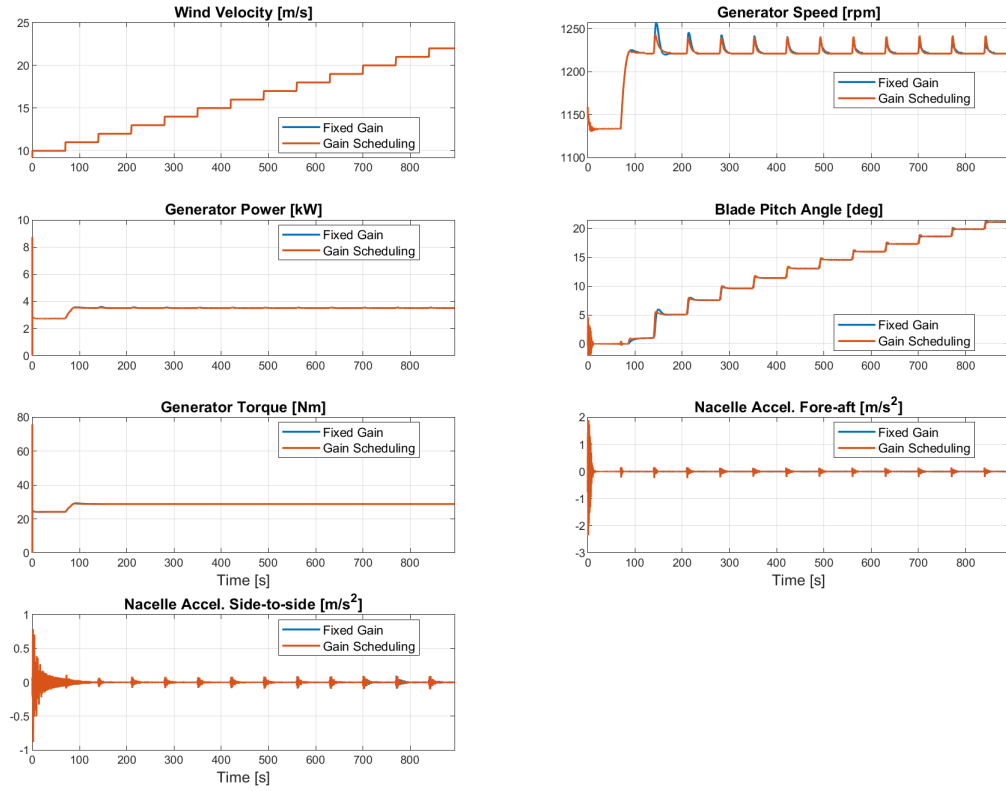


Figure 6.4: *Step response of the gain-scheduled controller and the fixed gain controller.*

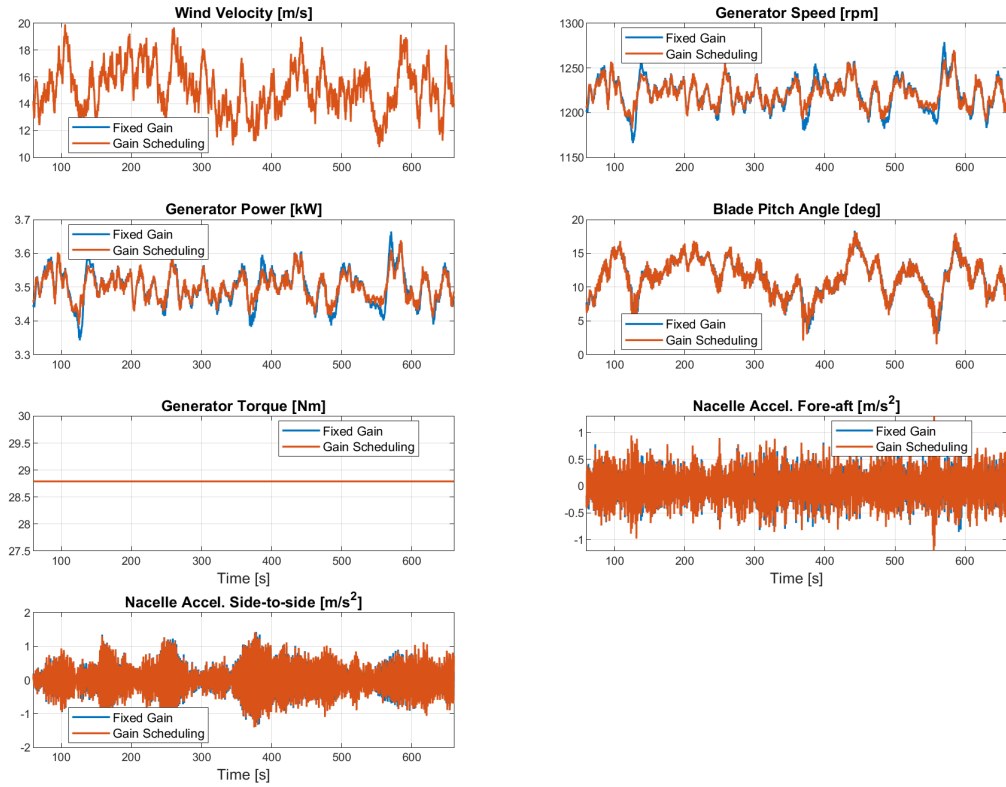


Figure 6.5: *System response to normal turbulence model at average wind speed of 15 [m/s] with the gain scheduled controller and the fixed-gain controller.*

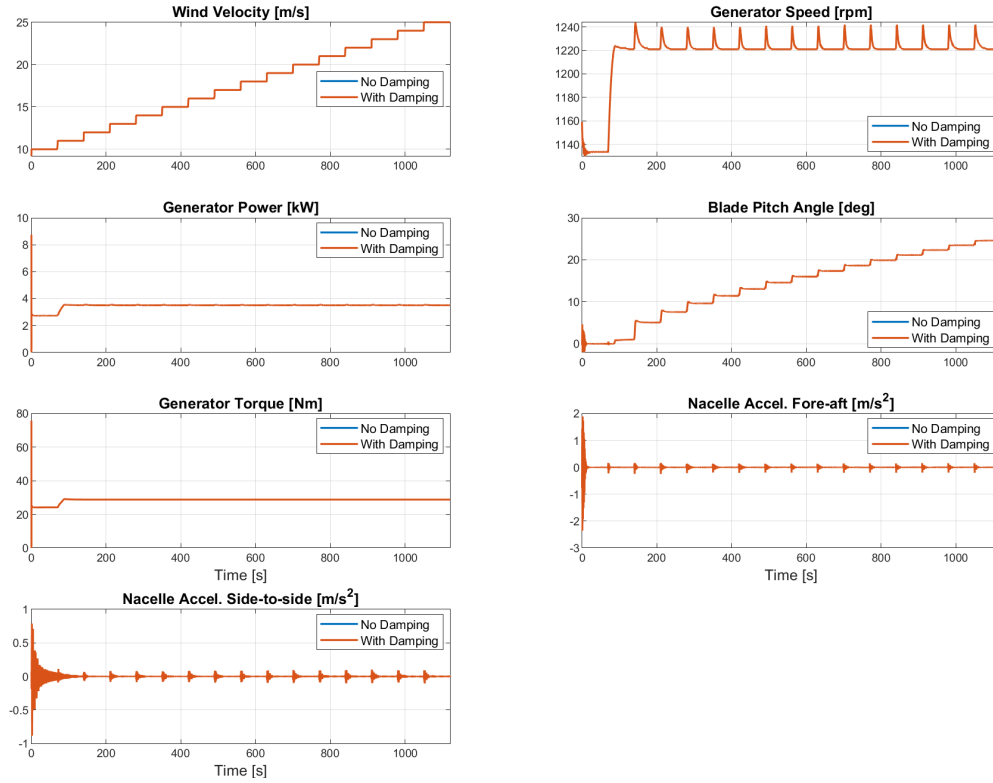


Figure 6.6: *Step response of the system with tower damping activated and no tower damping.*

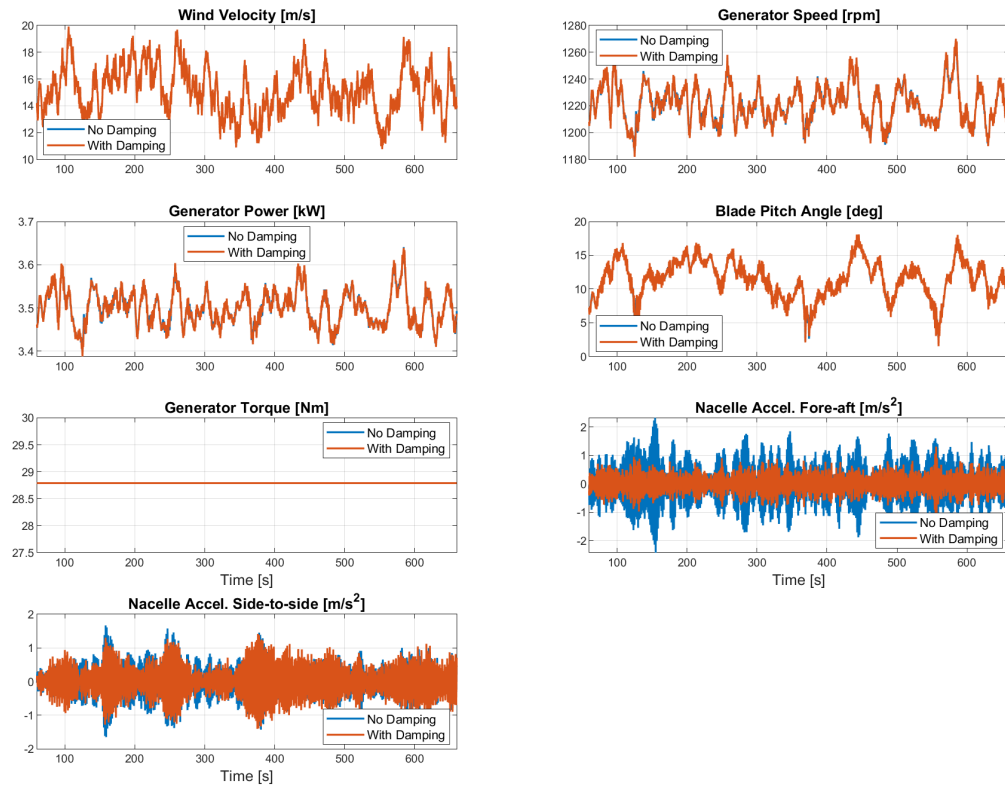


Figure 6.7: *System response to normal turbulence model at average windspeed 15 [m/s] with the tower damping activated and no tower damping.*

Chapter 7

Certification Analysis

7.1 Summary of the Analysis

Table 7.1: *Wind Turbine Class*

Parameter	Value
Wind speed Class	II
Turbulence Class	B

Each design load case (DLC) is evaluated using either fatigue (F) or ultimate (U) load analysis. Ultimate load cases are further categorized as normal (N), abnormal (A), which define the applicable safety factor γ_f . These classifications are summarized in [Table 7.2](#).

Table 7.2: *Design Load Case*

Design situation	DLC	Wind condition	Wind speeds	Type of analysis	Safety Factor
1) Power production	1.2	NTM	$V_{in} < V_{hub} < V_{out}$	F	-
	1.3	ETM	$V_{in} < V_{hub} < V_{out}$	U	N
	1.5	EWS	$V_{in} < V_{hub} < V_{out}$	U	N
5) Emergency shut down	5.1	NTM	$V_{hub} = V_r \pm 2 \text{ m/s and } V_{out}$	U	N
6) Parked (or idling)	6.1	EWM	-	U	N

Table 7.3: *Fatigue Damage Equivalent Loads for DLC 1.2*

Wind Speed (m/s)	D (10 min)	10-min Periods (25 y)
3	8.96e-15	77286
4	4.03e-15	102081
5	6.73e-15	121027
6	5.29e-14	132411
7	8.66e-14	135601
8	1.75e-12	131052
9	6.30e-12	120122
10	3.24e-12	104754
11	6.88e-12	87093
12	1.13e-11	69126
13	1.36e-11	52425
14	2.44e-11	38011
15	1.81e-11	26358
16	1.15e-11	17484
17	1.45e-11	11094
18	1.73e-11	6734
19	1.80e-11	3910
20	6.69e-11	2171
21	7.49e-11	1153
22	6.08e-11	585
23	3.39e-11	284
24	4.80e-11	132
25	4.86e-11	58
Total Damage (20 years):		5.68e-6

Table 7.4: *Ultimate Loads: % of Ultimate Characteristic Strength (UCS)*

DLC	Wind Speed (m/s)	% of UCS
1.3	7.0	12.88
	10.7	17.14
	17.0	17.34
1.5	7.0	8.66
	10.7	15.22
	17.0	11.41
5.1	8.7	12.89
	9.7	14.24
	10.7	15.11
	11.7	14.92
	12.7	16.31
	25.0	11.62
6.1	50.0	3.44

7.2 Detailed Analyses and Interpretation

7.2.1 Turbine IEC Class Determination

As discussed in [chapter 1](#), IEC Class II A/B has recently seen the highest number of wind turbines installed in onshore environments in Europe. This widespread adoption is largely due to the class's suitability for moderate wind conditions, which are common in many on-shore regions.

In [chapter 2](#), the selected sites for analysis, along with their corresponding Weibull distributions, align closely with the wind characteristics defined for IEC Class II B. As such, the turbine design process has been carried out with these parameters in mind from the start. Given this fact, it is both logical and efficient to perform certification under IEC Class II B specifications, ensuring the turbine meets the operational demands of the intended deployment environments.

7.2.2 Moments to Stresses

7.2.2.1 Blade Root Wall Thickness Estimation

The blade root is composed of triaxial glass fibre reinforced epoxy, a typical composite material used in wind turbine blades. The effective density of this material depends on the fibre volume fraction and the constituent material densities (glass fibres and epoxy resin). For typical manufacturing conditions and fibre volume fractions of 55–65%, the composite density falls within the range of 1900–2100 [kg/m³]. Based on this, a representative average value of $\rho = 2000$ [kg/m³] is used in this calculation [8].

The first meter of the blade, measured from the root, is known to be a near-perfect cylinder with uniform, straight walls. The required geometric and mass properties of this section are obtained from the results of [chapter 3](#), where both the root outer diameter and the local mass distribution are defined. Specifically, the outer diameter at the root is:

$$D_o = 3.16 \text{ [m]} \Rightarrow R_o = 1.58 \text{ [m]}$$

and the mass per unit length is:

$$m = 590 \text{ [kg/m]}$$

Given the cylindrical geometry and the known material density, the root section can be idealized as a hollow cylinder and thus the mass per unit length can be related to the annular cross-sectional area by:

$$m = \rho \cdot \pi (R_o^2 - R_i^2)$$

Rearranging and solving for the inner radius:

$$R_i^2 = R_o^2 - \frac{m}{\rho\pi} = 1.58^2 - \frac{590}{2000 \cdot \pi} \approx 2.4025 \Rightarrow R_i \approx 1.55 \text{ [m]}$$

The wall thickness is then calculated as:

$$t = R_o - R_i = 1.58 - 1.55 = \boxed{0.03 \text{ [m]} = 30 \text{ [mm]}}$$

Hence, the estimated blade root wall thickness is **30 [mm]**.

7.2.2.2 Area Moment of Inertia

The area moment of inertia is required to compute the shear stress resulting from torsional moments acting on the cylindrical blade root. For a hollow circular cross section, it is given by:

$$I = \frac{\pi}{64} (D_o^4 - D_i^4) = \frac{\pi}{64} (3.16^4 - 3.10^4) = 0.435 \text{ [m}^4\text{]}$$

where:

- $D_o = 3.16 \text{ [m]}$ is the outer diameter
- $D_i = 3.10 \text{ [m]}$ is the inner diameter,
- $I = 0.435 \text{ [m}^4\text{]}$ is the polar moment of inertia.

With the moment of inertia known, the shear stress at the outer surface of the root due to a given moment M (as obtained from FASTTool) is calculated as:

$$\sigma = \frac{M \cdot D_o/2}{I}$$

This relation assumes linear elastic behavior and that the maximum shear stress occurs at the outer fiber of the cylindrical section.

7.2.3 Turbine Behavior under Certification Load Cases

Before conducting Ultimate Limit State (ULS) and Fatigue Limit State (FLS) checks, the turbine's dynamic response is evaluated under Design Load Case (DLC) 1.2 at $V = 11$ [m/s]. This operating condition corresponds to the rated wind speed and is thus expected to result in high aerodynamic loading, potential structural resonance, and elevated fatigue stresses. Time-domain and frequency-domain responses are analysed below, focusing on blade root bending moments and tip deflection.

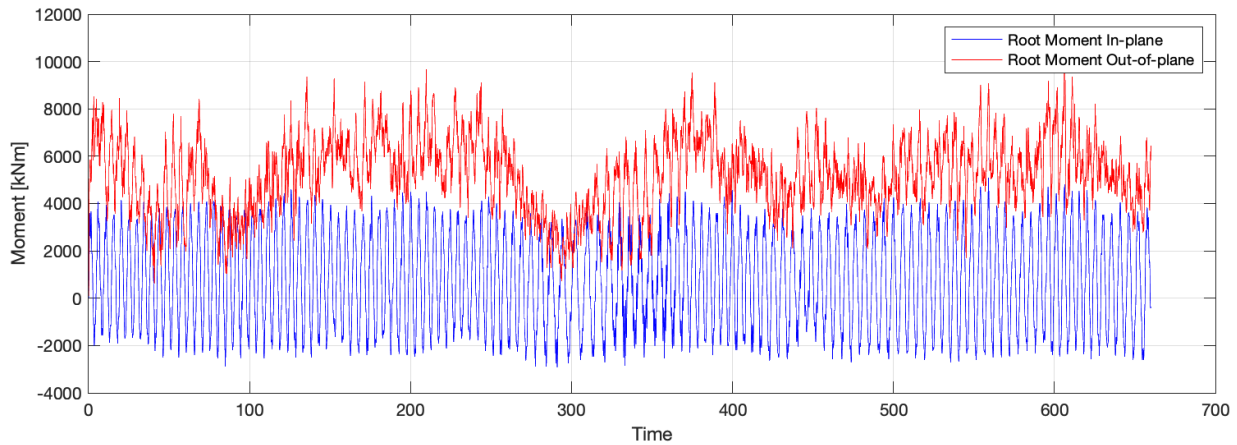


Figure 7.1: *Time series of in-plane and out-of-plane blade root bending moments under DLC 1.2 at 11 [m/s]. The out-of-plane moment dominates due to aerodynamic loading.*

Figure 7.1 shows the temporal evolution of the in-plane and out-of-plane root moments. As expected, the out-of-plane component exhibits significantly larger amplitude, dominated by cyclic aerodynamic loads. The signal shows periodicity frequency, indicating the influence of excitations due to rotor rotation and blade-tower interactions.

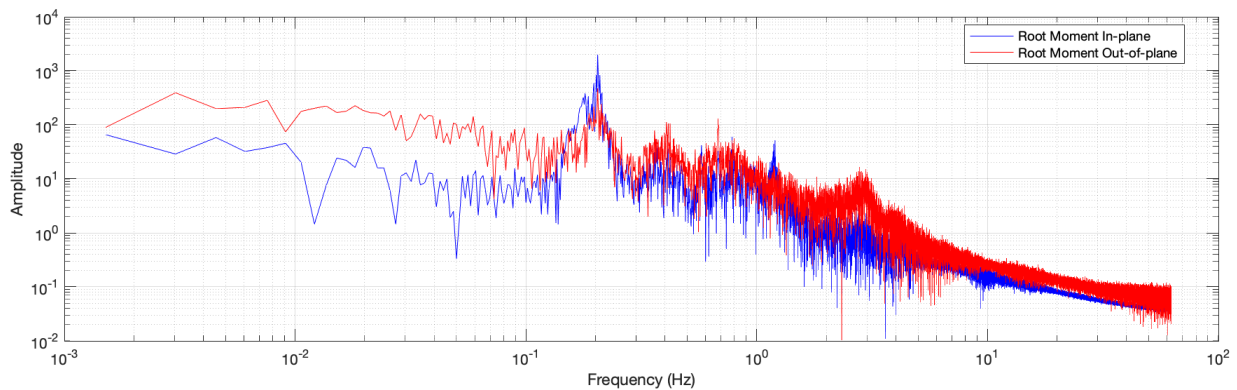


Figure 7.2: *Frequency spectra of blade root bending moments for DLC 1.2 at 11 [m/s].*

The spectral analysis in Figure 7.2 reveals peaks at approximately 0.2 [Hz], which corresponds to the frequency at start of operation, and 0.7 [Hz] and 1.2 [Hz], corresponding to the first flapwise and edgewise blade modes, respectively (see Table 5.8). This is concerning, as excitation at these frequencies indicates a risk of resonance, especially if the aerodynamic forcing (e.g., from 1P loading) aligns with these modes. Such alignment can amplify fatigue damage and dynamic stresses at the blade root.

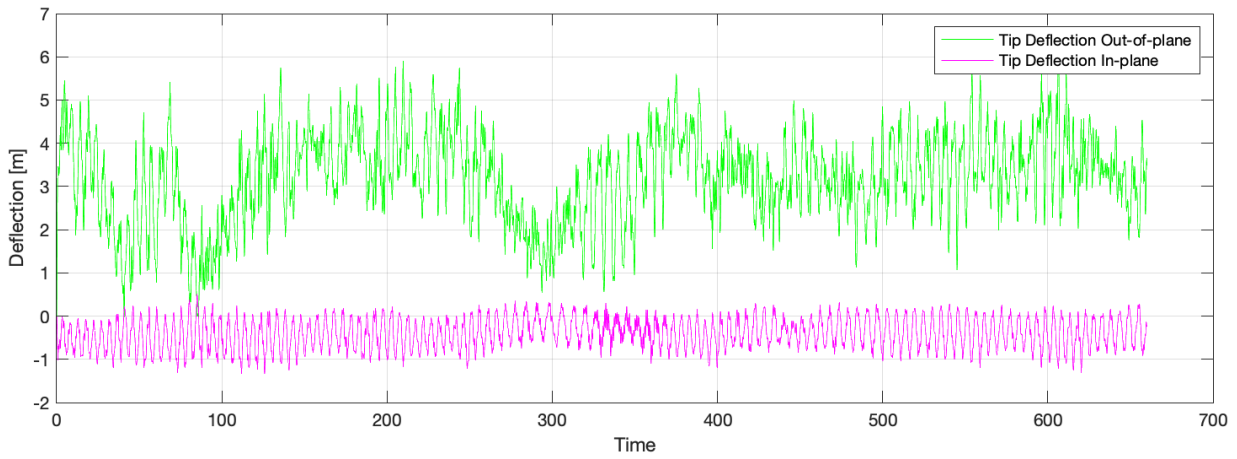


Figure 7.3: *Time series of blade tip deflection showing oscillatory behaviour for DLC 1.2 at 11 [m/s].*

Figure 7.3 displays the blade tip deflection over time. The response is governed by periodic oscillations, with amplitudes modulated by turbulence. The signal indicates that structural flexibility is a major factor in tip motion, requiring careful consideration in ULS checks for tower clearance and in fatigue analysis.

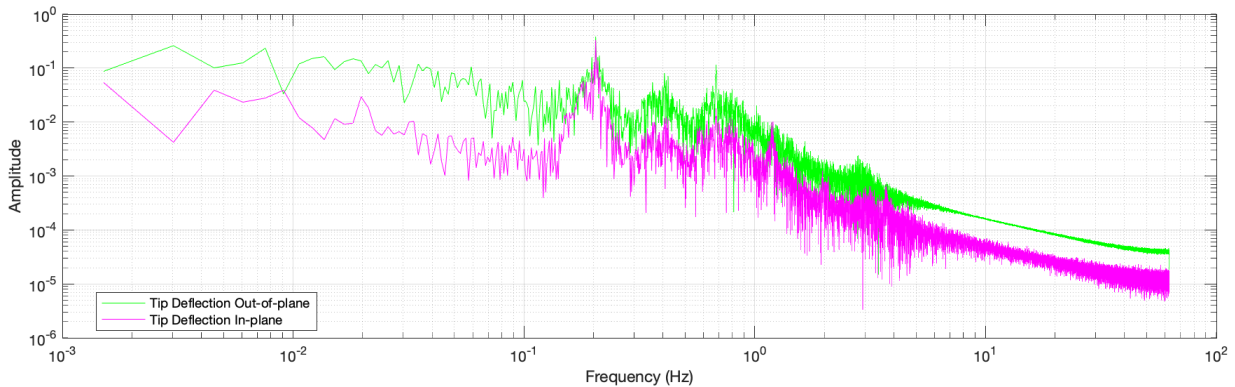


Figure 7.4: *Spectral content of tip deflection. A dominant peak near the blade's first natural flapwise frequency is observed for DLC 1.2 at 11 [m/s].*

The frequency content in Figure 7.4 confirms again peaks close to 0.2 [Hz], 0.7 [Hz] and 1.2 [Hz], with 0.7 [Hz] being the blade's first flapwise natural frequency. This suggests, as was described previously, that under rated conditions the rotor's 1P excitation interacts with this mode, without necessarily implying resonance.

The results confirm that DLC 1.2 at $V = 11$ [m/s] is a critical case for structural design due to the proximity of excitation frequencies to the blade's natural frequencies (see Table 5.8). While no resonance is definitively observed, the alignment with key modal frequencies necessitates caution in both fatigue and ultimate limit evaluations.

7.2.4 Selection of Partial Safety Factors for ULS and FLS

Table 7.5: Overview of Relevant Partial Safety Factors

Safety Factor	Symbol	Purpose	How to Choose It
Load	γ_f	Covers uncertainties in external loading	Based on load case type (N or A)
Material	γ_m	Covers uncertainties in material strength	Based on material type
Consequence of Failure	γ_n	Accounts for severity of component failure	Based on component class (1, 2, or 3)

The design and verification of the blade root strength under both ultimate and fatigue loading requires the application of appropriate partial safety factors (Table 7.5), as prescribed in certification standards such as the IEC 61400-1 [16]. The chosen safety factors account for uncertainties in loading, material properties, and the consequences of failure.

Ultimate Limit State (ULS):

- **Load factor** $\gamma_f = 1.35$: This value is prescribed for *normal* design situations according to IEC 61400-1 Table 3 [16], applicable for load case combinations without system faults or failures.
- **Material factor** $\gamma_m = 1.3$: As the blade root is constructed from glass fibre reinforced epoxy composite, failure from exceeding tensile or compressive strength is considered brittle and non-ductile. According to IEC 61400-1 Section 7.6.2.2, such components require a partial material factor of at least 1.3 to reflect the inherent variability and criticality of composite failure [16].
- **Consequence factor** $\gamma_n = 1.3$: Blade root failure is catastrophic and can cause the loss of a major component or the entire turbine. For such “non fail-safe” components, IEC 61400-1 assigns a consequence factor of $\gamma_n = 1.3$ (Component Class 3) [16].

Fatigue Limit State (FLS):

- **Load factor** $\gamma_f = 1.0$: Fatigue design load cases are based on long-term operational data, and IEC 61400-1 (Table 3) prescribes a unity safety factor for all fatigue simulations [16].
- **Material factor** $\gamma_m = 1.7$: Fatigue strength of composites is sensitive to degradation, manufacturing variation, and environmental effects. IEC 61400-1 recommends a conservative material factor of $\gamma_m = 1.7$ for fatigue damage assessment of composite materials [16].

- **Consequence factor** $\gamma_n = 1.3$: As in ULS, fatigue-induced failure at the blade root would be catastrophic. The same classification as Component Class 3 applies, requiring $\gamma_n = 1.3$ [16].

Table 7.6: Chosen Partial Safety Factors

Limit State	γ_f	γ_m	γ_n
Ultimate (ULS)	1.35	1.3	1.3
Fatigue (FLS)	1.00	1.7	1.3

All values presented in Table 7.6 are selected in compliance with the guidelines outlined in the IEC 61400-1 design standard [16], ensuring a consistent and conservative approach to structural verification of the blade root.

7.2.5 Ultimate Limit State Analysis

Using the estimated blade-root wall thickness, the converted moment-to-stress results from FASTTool, and the selected partial safety factors for the Normal operating case, the Ultimate Limit State (ULS) check is formulated as:

$$\gamma_f F_k \leq \frac{1}{\gamma_m \gamma_n} f_k \quad (7.1)$$

where:

- F_k is the stress result from the FASTTool using the moment-to-stress conversion,
- $f_k = 600$ [MPa] is the ultimate compressive strength (UCS) of the composite,
- γ_f , γ_m and γ_n are the partial safety factors for loading, material properties and consequence of failure, respectively (see Table 7.5).

Using this criterion, the applied stress as a percentage of UCS is computed for each Design Load Case (DLC). The resulting percentages are summarized in Table 7.4.

7.2.6 Fatigue Limit State Analysis

Fatigue in materials is a type of failure that occurs due to fluctuating stresses in a material over a large period of time. Determining the fatigue limit for a wind turbine is vital to ensure it will survive the design lifetime of 25 years. This load assessment is based on a simplification of the material, triax glass fibre reinforced epoxy used in the blade root. This composite material is non-isotropic, meaning the direction of the fibres in which the composite is arranged has a great impact on the stress it can experience. For this report, the material's ultimate

compressive strength is 600 [MPa], with an inverse slope of m of 9.

To determine the loads on a turbine, a 10-minute period is simulated in FastTOOL for wind speeds ranging from 3-25 [m/s]. Then, based on the Weibull distribution presented in [sub-section 2.2.1](#), the number of those 10-minute periods in 25 years is computed using the probability in a range of ± 1 [m/s]. This edgewise moment, simulated in FastTOOL is dominated by gravity and produces many irregular signals over the 10-minute period. To count the number of cycles in this signal, the Rainflow counting technique is implemented to provide the number of cycles and the range of loads experienced. With this, the damage for the given wind speed is computed using:

$$d_j = \frac{n_j}{N_j} \quad (7.2)$$

Repeating the procedure for the remaining wind speeds, the total damage is found with Miner's Rule, which needs to be less than one for the wind turbine to survive the loads experienced in 25 years.

$$D_{\text{total}} = \sum d_j \leq 1 \quad (7.3)$$

For 25 years, the total damage is found to be 0.00000568, validating the design from fatigue damage.

7.2.7 Mitigation of Tip Clearance Safety Factor

As seen in [Table 7.4](#), the turbine design meets all load certification requirements without issue. However, an increased tip deflection was observed, requiring corrective action. The new maximum tip deflection now measured at $\delta_{\text{max}} = 6.53$ [m], exceeds the [chapter 3](#) estimate of 5.13 [m]. The safety factor, defined as follows,

$$\text{SF} = \frac{C}{\delta} \geq 1.5$$

must be restored to avoid the blade from hitting the tower, where C is blade-tip clearance. Furthermore, as predicted in [chapter 5](#), the updated design for the tower has a larger radius and further reduces the available clearance.

Baseline Maximum Tip Deflection from Task 3 (Nacelle Tilt = 3°)

$$C_{3^\circ} = 7.66 \text{ [m]}, \quad \delta_{\text{est}} = 5.13 \text{ [m]} \quad \longrightarrow \quad \text{SF}_3 = \frac{7.66}{5.13} = 1.49 \approx 1.5.$$

Actual Maximum Tip Deflection of New Design

$$\delta_{\text{act}} = 6.53 \text{ [m]} \longrightarrow \text{SF}_3 = \frac{7.66}{6.53} = 1.17 < 1.5.$$

The new maximum tip deflection does not meet the tip clearance requirements. Therefore, two corrective actions are considered:

1. Increase blade thickness by 5%

Scaling the laminate thickness factor from $k = 1.00$ to $k = 1.05$ stiffens the blade, reducing the maximum tip deflection to

$$\delta_{k=1.05} \approx 4.98 \text{ [m]},$$

hence

$$\text{SF}_{k=1.05} = \frac{7.66}{4.98} = 1.54 > 1.5.$$

However, material usage grows as k increases, significantly increasing raw-material and manufacturing costs.

2. Increase nacelle tilt to 5.5°

Raising the tilt from 3° to 5.5° increases static clearance to

$$C_{5.5^\circ} \approx 10.2 \text{ [m]},$$

yielding

$$\text{SF}_{5.5^\circ} = \frac{10.2}{6.53} \approx 1.56 > 1.5.$$

The only penalty from this approach is a modest power reduction ($\approx 1\%$) due to the reduced normal wind component.

Selected Solution The nacelle tilt adjustment to 5.5° is adopted, as it achieves the required safety factor with minimal performance loss and negligible capital cost.

Chapter 8

Reflection and Prospect

8.1 Reflection

The process of designing a 3.5 [MW] onshore wind turbine aimed to convert the reference NREL 5 [MW] turbine into a smaller, cost-effective alternative. Preliminary aerodynamic analyses identified optimal pitch angles that maximise the power coefficient C_P for the chosen design ratio. A comparison between the expected power curve from [subsection 2.2.3](#) and the actual power curve obtained after detailed generator and aerodynamic design in [subsection 5.2.2](#) demonstrates that the final design meets the predefined objectives from a power production standpoint. The behaviour of the turbine is displayed in [Figure 8.1](#) and follows the same trends when compared to the NREL5MW design for all shown parameters [\[19\]](#). The disparity worth noting the most is that the final design has a slimmer Region 2½ due to the reasons shown in [subsubsection 5.2.1.2](#).

From a validation perspective, the design appears to be overconstrained, as the computed stress values are significantly lower than the allowable limits. This suggests that the structure could be optimised further, for instance, by using a lower thickness factor. However, the blade tip deflection remains a concern, reaching nearly 28%, which exceeds expectations. While the reference tip deflection is 5.5 [m], and the expected value for this design was 5.1 [m], the final output is 6.53 [m]. To address this issue, several potential solutions can be considered, including pre-bending the blade, introducing a cone angle, or modifying the blade geometry through alternative airfoil profiles or increased twist near the tip.

In summary, to ensure the successful continuation of the wind turbine development, it is recommended to optimise the structural design to reduce unnecessary mass, while addressing excessive blade tip deflection, and validating performance to meet both safety and cost-effectiveness targets.

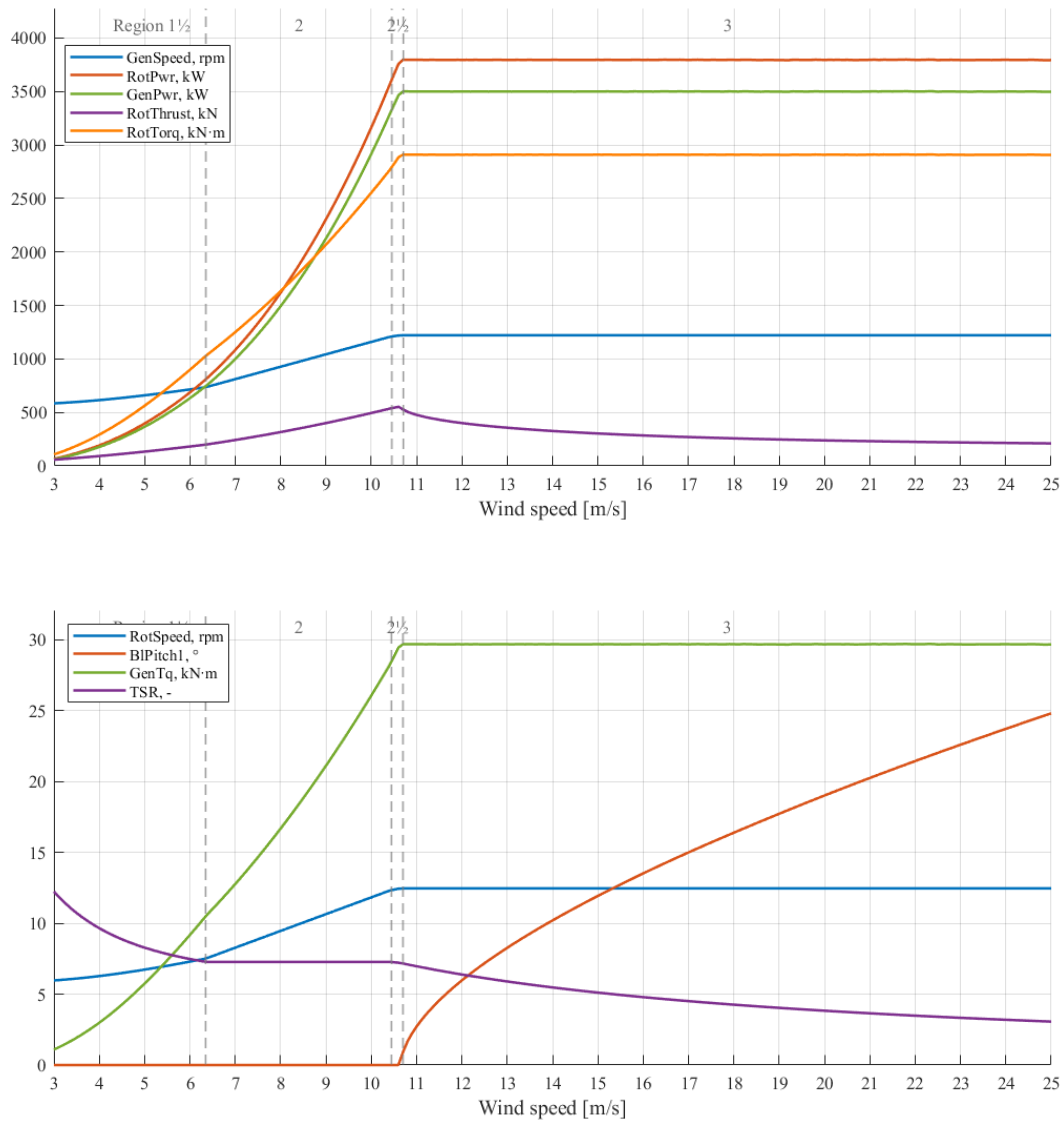


Figure 8.1: *Steady-state responses as a function of wind speed of the final design*

8.2 Prospect

A reasonable way to assess the overall viability of the proposed design is to observe how energy production and manufacturing costs have scaled with respect to the NREL5MW turbine. For this purpose, turbine masses have been used as a proxy for material costs to approximate capital expenditure differences between the two designs, as seen in [Table 8.1](#). Comparing the costs of both designs reveals that there has been an overall cost reduction of around 100,000€ or 21%.

While mass is an imperfect substitute for cost, especially without more detailed component-level analysis, the AEP-to-mass ratio offers a useful indicator of the financial implications of scaling. If the energy yield increase outweighs the incremental mass (and presumed cost),

Table 8.1: *Material cost comparison between design and reference turbines*

	Design turbine		Reference Turbine		
Component	Mass [Kg]	Cost [€]	Mass [Kg]	Cost [€]	Price per kg [€]
Tower	376,635	≈ 240,000	347,460	≈ 218,500	0.63 [2]
Blades	3 × 14,387.5	≈ 64,743	3 × 17,740	≈ 79,828	1.5[1]
Nacelle	125,981	≈ 80,000	296,780	≈ 186,971	0.63 [2]
Total Cost		≈ 384,743		≈ 485,299	

the proposed turbine may offer a viable alternative and continuation of the development may be encouraged. The estimated AEP of the reference turbine is around 18.378 [GWh/year], while the proposed design is 22% lower, producing 14.349 [GWh/year]. This means that the expected cost reduction in the design matches the decrease in productivity (within 1%), resulting in a product that retains a similar return on investment as the NREL5MW design. This suggests that the design produced in this report is a competitive product and thus further development is encouraged.

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